AOS-CX 10.15.xxxx Monitoring Guide

5420, 6200 Switch Series

Hewlett Packard Enterprise

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Copyright Information

Contents

Chapter 1

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- HPE Aruba Networking 5420 Switch Series (S0U67A, S0U55A, S0U63A, S0U64A, S0U65A, S0U75A, S0U72A, S0U78A, S0U58A, S0U73A, S0U74A, S0U71A, S0U76A, S0U70A, S0U77A, S0U60A, S0U61A, S0U62A, S0U66A, S0U68A)
- Aruba 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A, R8Q67A, R8Q68A, R8Q69A, R8Q70A, R8Q71A, R8V08A, R8V09A, R8V10A, R8V11A, R8V12A, R8Q72A, JL724B, JL725B, JL726B, JL727B, JL728B, S0M81A, S0M82A, S0M83A, S0M84A, S0M85A, S0M86A, S0M87A, S0M88A, S0M89A, S0M90A, S0G13A, S0G14A, S0G15A, S0G16A, S0G17A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

Command syntax notation conventions

<table>
<tr><th>Convention</th><th>Usage</th></tr>
<tr><td><code>example-text</code></td><td>Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([ ]).</td></tr>
<tr><td>example-text</td><td>In code and screen examples, indicates text entered by a user.</td></tr>
<tr><td>Any of the following:
<code><example-text></code>
<code><example-text></code>
<code>example-text</code>
example-text</td><td>Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code:
For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value.
For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.</td></tr>
<tr><td><code>|</code></td><td>Vertical bar. A logical OR that separates multiple items from which you can</td></tr>
</table>

Convention	Usage
	choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
`{ }`	Braces. Indicates that at least one of the enclosed items is required.
`[ ]`	Brackets. Indicates that the enclosed item or items are optional.
… or `. . .`	Ellipsis: ▪ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ▪ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

`switch>`

Indicates the operator command context.

`switch#`

Indicates the manager command context.

`switch(CONTEXT-NAME)#`

Indicates the configuration context for a feature. For example:

`switch(config-if)#`

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

`switch(config-vlan-<VLAN-ID>)#`

Where *<VLAN-ID>* is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: ***member/slot/port***.

On the Aruba 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

On the Aruba 6400 and 5420 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/1 and 1/2.
  - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

Chapter 2

Monitoring hardware through visual

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.

For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the Aruba Switch Documentation section of the Aruba Hardware Documentation and Translations Portal.

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.

   On the front of the switch, verify that the states of the following LEDs are On Green:
   - Power
   - Health
2. Verify that the Health LEDs of all installed line modules are On Green.
3. Verify that the Health LEDs of all installed management modules are On Green.
4. Verify that the network ports are operating normally.
   a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:
      - On Green (normal operation)
      - Off (no line module installed)

   b. On each line module, verify that each port LED is in one of the following states:
      - On Green, Half-Bright Green, or Flickering Green (normal operation)
      - Off (no cable connected or port off by default in config)
5. Verify that the power supplies are operating normally.
   a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:
      - On Green (normal operation)
      - Off (no power supply installed)

   b. On each power supply, verify that LEDs are in the following states:
      - Power LED: On Green
      - Fault LED: Off
6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:
   a. Verify that the LEDs for the fabric modules are in one of the following states:
      - On Green (normal operation)
      - Off (component not installed)

   b. Verify that the LEDs for the fan trays and fans are On Green.

7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:
   - Health LED is On Green.
   - Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.

Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails. For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the Aruba Switch Documentation section of the Aruba Hardware Documentation and Translations Portal.

Procedure

1. Detect if the standby management module is shut down.

   If the standby management module is shut down, the LED states are as follows:
   - The standby management module health LED is Off.
   - The standby management state active (Actv) LED is Off.
   - The standby management state standby (Stby) LED is Off.
   - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.
2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:
   - The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
   - The standby management module Booting LED is Slow Flash Green when the AOS-CX operating system is booting.
   - The standby management state active (Actv) LED is Off.
   - The standby management state standby (Stby) LED is Off.
   - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.
3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:
   - On the active management module, in the Status Rear section, the LED for the fabric module is Off.
   - On the rear display module, the LED for the fabric module is Off.
   - On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.

All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.). For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the Aruba Switch Documentation section of the Aruba Hardware Documentation and Translations Portal.

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.

   The chassis health LED is located on the front of the switch and on the rear panel of the switch.
2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:

   The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

   If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.
3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:
   - The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
   - The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.
4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.

   The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.
5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state:

   Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

Chapter 3
IP Flow Information Export

IP Flow Information Export (IPFIX) is an embedded network flow analysis tool that compiles characteristic and measured properties of flows and sends flow reports to internal or external flow collectors. IPFIX is configurable via the command-line or REST interfaces. With IPFIX, customers configure flow records with match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports.

Compatibility with Traffic Insight

The AOS-CX traffic insight feature allows monitoring of large amount of data that it collects from various flow exporters like IPFIX, and provides the ability to filter, aggregate, and sort the data based on user flow monitor requests. Traffic insight tracks different monitor requests simultaneously and provides monitor reports per request. For more information on configuring the Traffic Insight features, refer to the *AOS-CX Security Guide*.

Information Elements

The IPFIX Information Elements (IE) are entities that are defined and maintained by the Internet Assigned Numbers Authority (IANA). They are characterized by a unique piece of information they can provided about a flow. Information Elements may be either private or public. Private Information Elements are exported with a Private Enterprise Number (PEN).

AOS-CX can act as an intermediate collecting process for flow reports from hardware to append certain additional IPFIX information elements to the flow reports. When configured, the software will act as an intermediate exporting process to export the augmented flow reports to any configured flow exporters.

Standard Information Elements
(N/A)

Private Information Elements
tcp3WayHsServerRespTime (ID 1101, 4823 Aruba)
tcp3WayHsClientRespTime (ID 1102, 4823 Aruba)

Flow monitors

A flow monitor is applied to an interface to perform network traffic monitoring. A flow monitor consists of a flow record, a flow cache, and optional flow exporters. A flow record must be created and assigned to the flow monitor for the monitoring process to function. Flow data is compiled from the network traffic on the interface and stored in the flow cache based on the match (key) and collect (non-key) fields in the flow record. Data from the flow cache is exported by the flow exporters assigned to the flow monitor. 6200 series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor.

Flow Records

A flow record defines match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters. On the 6200, 6300, 6400, 8100, and 8360 switch series a maximum of sixteen flow records can be created.

There are six mandatory match fields, of which the IP match fields must be of the same type (IPv4 or IPv6).

A flow record is invalid if it does not contain one of the supported sets of match fields.

The supported sets of match fields are:

1. IPv4:

- IPv4 version
- IPV4 source address
- IPv4 destination address
- IPv4 protocol
- Transport destination port
- Transport source port

2. IPv6:

- IPv6 version
- IPv6 source address
- IPv6 destination address
- IPv6 protocol
- Transport destination port
- Transport source port

Destinations

The destination specifies where flow reports are sent. There are two possible types of destination for a flow exporter:

1. (default) Hostname or IP address of a device with an optional VRF
2. Traffic Insight instance

A flow exporter can only send flow reports to one destination. The destination type specifies which destination to use. If no destination type is specified, the default destination type is the hostname or IP address of a device with an optional VRF. (If a VRF is not specified, the default VRF will be used.) A destination of each type can be configured, but only the one corresponding to the destination type is used. If there is no destination corresponding to the destination type, then the flow exporter configuration is incomplete. If a new destination of a particular type is configured, it will replace the destination of that type that was previously configured.

Configuring IP Flow Information Export on 6200 Switches

The following list describes the steps required to configure a IP flow information export (IPFIX) solution:

- Step one: Create flow records
- Step two: Configure flow exporter(s)
- Step three: Configure monitor(s)
- Step four: Apply a flow monitors to interface(s)

IPv6 related commands are only applicable to switches that support IPv6 protocol.

Step one: Create Flow Records

Flow Records are used to define the data that will be added to the IPFIX template. This example configures one record for IPv4 and one for IPv6.

```
switch(config)# flow record flowRecordv4
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
switch(config-flow-record)# collect application https url

switch(config)# flow record flowRecordv6
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
switch(config-flow-record)# collect application https url
```

Next, use the **show flow record** command to verify the configuration.

```
--------------------------------------------------------------
Flow record 'flowRecordv4'
---------------------------------------------------------------------------------
Description              : ipv4
Status                   : Accepted
Match Fields
```

```
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    ipv4 version
    transport destination port
    transport source port
Collect Fields
    application name
    counter bytes
    counter packets
    application https url
    timestamp absolute first
    timestamp absolute last

--------------------------------------------------------------------------------
Flow record 'flowRecordv6'

--------------------------------------------------------------------------------
Description              : ipv6
Status                   : Accepted
Match Fields
    ipv6 destination address
    ipv6 protocol
    ipv6 source address
    ipv6 version
    transport destination port
    transport source port
Collect Fields
    application name
    counter bytes
    counter packets
    application https url
    timestamp absolute first
    timestamp absolute last
```

Step two: Configure flow exporter(s)

In this step, you can define an exporter to send to an external destination by hostname or IP address, or to an internal destination such as Traffic Insight. The example below configures IPFIX to export data to an external address/hostname:

```
switch(config)# flow exporter flowExternal
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 11.1.1.1
switch(config-flow-exporter)# show flow exporter
--------------------------------------------------------------------------------
Flow exporter 'flowExternal
--------------------------------------------------------------------------------
Status                  : Accepted
Export Protocol         : ipfix
Destination Type        : Hostname or IP address
Destination             : 11.1.1.1
Transport Configuration
Protocol            : udp
Port                : 4739
```

To configure IPFIX to export to Traffic Insight, first configure Traffic Insight.

```
switch(config)# traffic-insight TI
switch(config-ti-TI)# source ipfix
switch(config-ti-TI)# monitor topN type topN-flows
switch(config-ti-TI)# monitor raw-mon type raw-flows
switch(config-ti-TI)# enable
```

Next, configure the flow exporter for Traffic Insight

```
switch(config)# flow exporter flowExpTI
switch(config-flow-exporter)# export-protocol ipfix
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight TI
```

You can use the **show flow exporter** command to verify the flow exporter configuration for Traffic Insight

```
switch(config)# show flow exporter flowExpTI

--------------------------------------------------------------------------------
Flow exporter 'flowExpTI'
--------------------------------------------------------------------------------
Status                  : Accepted
Export Protocol         : ipfix
Destination Type        : Traffic Insight
Destination             : TI
Transport Configuration
Protocol            : udp
Port                : 4739
```

Finally, use the **show run traffic-insight** command to verify the Traffic Insight configuration:

```
switch(config)# show running-config traffic-insight
traffic-insight TI
enable
source ipfix
!
monitor topN type topN-flows entries 5
monitor appFlow type application-flows
monitor raw type raw-flows
```

Step three: Configure the monitor(s)

First, configure an IPv4 flow monitor.

```
switch(config)# flow monitor flowMonv4
switch(config-flow-monitor)# record flowRecordv4
Switch (config-flow-monitor)# exporter flowExternal
switch(config-flow-monitor)# exit
```

Next, configure an IPv6 flow monitor.

```
switch(config)# flow monitor flowMonv6
```

```
switch(config-flow-monitor)# record flowRecordv6
switch(config-flow-monitor)# exporter flowExternal
switch(config-flow-monitor)# exit
```

Once both flow monitors are created, use the **show flow monitor** command to verify the flow monitor configurations.

```
switch(config-flow-monitor)# show flow monitor
--------------------------------------------------------------------------------
Flow monitor 'flowMonv4'
--------------------------------------------------------------------------------
Status                    : Accepted
Flow Record               : flowRecordv4
Flow Exporter(s)          : flowExternal
Cache Configuration
Inactive Timeout      : 30
Active Timeout        : 1800

--------------------------------------------------------------------------------
Flow monitor 'flowMonv6'
--------------------------------------------------------------------------------
Status                    : Accepted
Flow Record               : flowRecordv6
Flow Exporter(s)          : flowExternal
Cache Configuration
Inactive Timeout      : 30
```

Role-based IPFIX

If IPFIX monitoring is configured on a port, a network administrator who wants to monitor traffic from clients must preconfigure IPFIX monitor before onboarding the clients. Also, if a client moves from one port to another, the monitor must be reconfigured accordingly. To reduce all these complexities, 6200, 6300 and 6400 switch series allow you to configure an IPFIX monitor on a port access role.

An IPFIX deployment with monitoring configured on a port uses TCAM to get flow statistics. Since TCAM is a high-demand resource which is shared across multiple protocols including the security protocols, security policy applications could fail if IPFIX consumes most of the TCAM resources. Port access clients can use a large amount of of TCAM resources for policy applications, making it difficult support port access clients with TCAM based IPFIX.

With role-based IPFIX, when a user plugs a device into to a specific port, their user role is applied to the port. Whenever the user plugs that device into to a different port again, the role is applied to that specific port. This gives mobility and accessibility to the user and provides permissions specific to the user's role when the user's device moves across ports. Role-based IPFIX enables colorless port behavior.

There are some mutually exclusive features that will disable role-based IPFIX when they are enabled. The mutually exclusive features are:

- UBT
- MAC Lockout
- Extended Router MAC
- Source Port Filter
- IP Lockdown
- L3VNI

The following behaviors are observed in a deployment with role-based IPFIX:

- If a user applies the IPFIX monitor configuration on the port, it uses the existing TCAM infrastructure.
- If a port has both Role based IPFIX and Port based IPFIX, the Port based IPFIX implementation takes the priority.
- If IPFIX monitor configuration on a port moves from role based to TCAM based or vice-versa, existing IPFIX flows learned on that port are deleted.
- When the user role has an IPFIX configuration and fails to apply it, port access logs it, but it does not prevent the client from getting authorized.
- If port access is in the client mode and if there is more than one client with a conflicting IPFIX monitor configuration, traffic from all the clients will be exported through the IPFIX monitor configured for the first client.
- Role-based IPFIX and TCAM port-based IPFIX implementations can co-exist on a switch.
- Flows from unauthorized clients are not monitored.

These are some limitations to role-based IPFIX:

- Role based IPFIX can be enabled only through port access role assignment.
- Only Ingress flow monitoring is supported on both role-based and port-based IPFIX.

FAQs and Troubleshooting

- The following messages are displayed to indicate an illegal argument:
  - % The flow exporter <EXPORTER-NAME> does not exist.
  - % The flow record <RECORD-NAME> does not exist.
  - % The flow monitor <MONITOR-NAME> does not exist.
  - Invalid destination IP address or hostname entered.
  - Unable to create the flow exporter. The maximum allowed number of flow exporters (<max>) has been reached.
  - Unable to create the flow record. The maximum allowed number of flow records (<max>) has been reached.
  - Unable to create the flow monitor. The maximum allowed number of flow monitors (<max>) has been reached.
  - Flow monitor cannot be applied while interface is part of LAG <LAG-NAME>.
  - Flow monitor could not be applied.
  - Flow monitor could not be unapplied

Flow monitoring commands

collect application tcp establishment-time

```
collect application tcp establishment-time
[no] collect application tcp establishment-time
```

Description

Configures collect (non-key) fields for a flow record when in the **config-flow-record** context.

Supported on IPv4 and IPv6 flows.

Only one collect field can be specified per line, while a flow record can have multiple collect fields.

Parameter	Description
`tcp`	Specifies Transmission Control Protocol (TCP) parameters as a non-key field in a flow record.
`establishment-time`	Specifies TCP connection establishment time as a non-key field in a flow record.

Usage

- ARC inspects the first few packets of each flow and only those packets are copied to the switch.
- Time measurement for established TCP flows is possible during the TCP connection establishment phase, and all three packets are received in the order.

Examples

Adding a TCP connection establishment time to flow record **flow-record-1** as a collect field.

```
switch(config-flow-record)# collect application tcp establishment-time
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15.1000	Command introduced

Command Information

Platforms	Command context	Authority
6200	`config` `config-flow-collector`	Administrators or local user group members with execution rights for this command.

diag-dump ipfix basic

`diag-dump ipfix basic`

Description

Displays diagnostic information for IPFIX.

Examples

```
diag-dump ipfix basic
==============================================================================
```

```
[Start] Feature ipfix Time : Tue Apr 11 02:23:03 2023
==========================================================================
--------------------------------------------------------------------------
[Start] Daemon ipfixd
--------------------------------------------------------------------------
- IPFIX Record Cache dump -
- IPFIX Record ipfix -

....

:- IPFIX Monitor v6ti completed -
- End of IPFIX Monitor Cache dump -
--------------------------------------------------------------------------
[End] Daemon ipfixd
--------------------------------------------------------------------------
--------------------------------------------------------------------------
[Start] Daemon ops-switchd
--------------------------------------------------------------------------
Key format: <traffic_type>_<coalescence_id>_<agent_id>_<asic_port>
Key                               TCAM Entry ID    Count
--------------------------------- ---------------- -----
1_1532781829_3_20                 0xffff7c7e7a00   1
1_3217499901_1_12                 0xffff91187580   1
1_3217499901_1_13                 0xffff91183d80   1
1_3217499901_1_14                 0xffff91186e80   1

....

--------------------------------------------------------------------------
[End] Daemon ops-switchd
--------------------------------------------------------------------------
==========================================================================
[End] Feature ipfix
==========================================================================
Diagnostic-dump captured for feature ipfix
```

For more information on features that use this command, refer to the Security Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced on 9300S and 6200 Switch series

Command Information

Platforms	Command context	Authority
6200	Manager (#)	Administrators or local user group members with execution rights for this command.

flow exporter

```
flow exporter <name>
  export-protocol ipfix
  description <description>
  destination
```

```
    <hostname> [vrf vrfname]
    <ipaddr> [vrf vrfname]
    <ip6addr>  [vrf vrfname]
    type {hostname-or-ip-addr |
 no ..
 template data timeout <timeout>
 transport udp <port>
```

Description

A flow exporter is the part of the IP Flow Information Export (IPFIX) feature that defines how a flow monitor exports flow reports. You can assign the same flow exporter configuration to more than one flow monitor. Each flow exporter includes a destination setting that identifies the device to which the flow reports are sent. series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor.

Parameter	Description
`<name>`	Name of the flow exporter, up to 64 characters.
`description <description>`	A description of the flow exporter, up to 256 characters and spaces.
`destination  <hostname>|<IPaddr>|<ip6addr>`	
`[vrf vrfname]`	You can optionally include the name of the destination VRF in the destination definition.
`destination type {hostname-or-ip-addr | traffic-insight}`	The exporter sends flow reports to a traffic insight destination.
`destination traffic-insight <name>`	The exporter sends flow reports to a specific traffic insight destination.
`no ..`	Negate any configured parameter.
`template data timeout <timeout>`	A flow exporter template describes the format of exported flow reports. Therefore, flow reports cannot be decoded properly without the corresponding templates. This setting defines how often the flow exporter will resend templates to the flow monitor. The supported range is 1-86400 seconds, and the default is 600 seconds.
`transport udp <port>`	Transport protocol and port for sending flow record reports. The default port is port 4739,
`type`	Configure the type of the destination.
`IPV4-ADDR`	The IPv4 address of the

Parameter	Description
	destination for the flow exporter.
INSTANCE-NAME	Che name of the Traffic Insight instance.

Examples

The following example creates a flow exporter configuration named **exporter-1**.

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# dscp 34
switch(config-flow-exporter)# destination 192.0.2.1 vrf VRF1
switch(config-flow-exporter)# template data timeout 1200
switch(config-flow-exporter)# description Exports flows to 192.0.2.1
```

The following example sets a Traffic Insight instance as the destination for a flow exporter

```
switch(config)# flow exporter exporter-3
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight instance-1
```

Following example adds a destination of each possible type and set **hostname-or-ip-addr** as the type to use:

```
switch(config)# flow exporter exporter-4
switch(config-flow-exporter)# destination collector-1
switch(config-flow-exporter)# destination traffic-insight instance-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
```

Following example removes the destination of type **hostname-or-ip-addr** from a flow exporter

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# no destination
```

Following example removes the destination type from a flow exporter

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# no destination type
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced on 9300S and 6200 Switch series

Command Information

Platforms	Command context	Authority
6200	`config` `config-flow-exporter`	Administrators or local user group members with execution rights for this command.

flow monitor

```
flow monitor <name>
   exporter <name>
   cache timeout {inactive <timeout} |{active <timeout}
   description <description>
```

Description

If no software augmentation of flows is required, there is no need to configure a flow collector or flow monitor.

Parameter	Description
`<name>`	Name of the flow monitor, up to 64 characters.
`cache timeout active`	Use the cache timeout parameter to define an active or inactive timeout for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer than the inactive timeout. The active timeout range is 30-604800. The default active time out value is 1800 and inactive timeout value is 30.
`cache timeout`	Use the cache timeout parameter to define an inactive timeout for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer than the inactive timeout.
`exporter <name>`	Assign a flow exporter to a flow monitor.

Examples

Add or modify the description of flow exporter **flow-exporter-1**

```
switch(config)# flow exporter flow-exporter-1
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced on 9300S and 6200 Switch series

Command Information

Platforms	Command context	Authority
6200	`config` `config-flow-monitor`	Administrators or local user group members with execution rights for this command.

flow record

```
flow record <name>
  match
    ipv6 {protocol|version}|{source|destination address}
    ip {source | destination} address
    ip {protocol | version
    transport {source|destination} port
    timestamp absolute first
    timestamp absolute last
  collect
    application name
    application https url
    counter {packets|bytes}
  description <description>
```

Description

Define data to be included in a flow record by configuring flow record match and collect fields.

A flow record defines match (key) fields and collection (non-key) fields. Customers configure flow records with **match** (key) fields and **collect** (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collect fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

Traffic with matching attributes (for example, traffic coming from the same interface, sent to the same destination with the same protocol) are classified as a single flow. Information for some or all of the matched settings can be collected and exported to a destination defined by the flow exporter assigned to the flow monitor.

Traffic must match a match rule definition before it can be collected and sent. You cannot collect and send data that is not matched.

Parameter	Description
`<name>`	Name of the flow monitor, up to 64 characters.
`match`	match traffic according to one or more of the following key attributes: ■ **ip**: match traffic on an IPv4 network ■ **protocol**: Match traffic using the same IP protocol ■ **source**: Match traffic from the same source ■ **destination**: Match traffic to the same destination ■ **address**: Match traffic by source or destination IP address ■ **transport**: Match traffic by source or destination transport type ■ **port**: Match traffic by source or destination transport port
`description`	A description for the flow record up to 256 characters long, including spaces.
`collect`	Configures data fields to be included a flow record. ■ **application name:** Specify the application name as a non-key field in a flow record.

Parameter	Description
	■ **application https url**: Specify the HTTP/HTTPS application URL as a non-key field in a flow record. ■ **application tcp establishment-time**: Specifies TCP connection establishment time as a non-key field in a flow record ■ **counter packets**: Collect counter data for packets in the flow. Packet count represents the number of incoming packets since the previous report. ■ **counter bytes**: Collect counter data for bytes in the flow. Byte count represents the number of incoming bytes since the previous report. ■ **timestamp absolute first**: Collect absolute timestamp of the first packet observed.

Examples

Adding IPv4 and transport match fields to **flow-record-1:**

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport source port
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# description Record used for basic ipv4 traffic
analysis
```

Adding counter and timestamp collect fields to **flow-record-1**:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
```

Add IPv4 match fields to flow record **flow-record-1** using the `ip` keyword

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
```

Remove the IPv4 destination address match field from flow record **flow-record-1** using the `ip` keyword

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ip destination address
```

Remove the IPv4 destination address match field from flow record **flow-record-1** using the `ipv4` keyword

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ipv4 destination address
```

Remove the transport destination port match field from flow record **flow-record-1**

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match transport destination port
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced on 6200 and 9300S Switch series

Command Information

Platforms	Command context	Authority
6200	`config` `config-flow-record`	Administrators or local user group members with execution rights for this command.

flow-tracking

```
flow-tracking
  enable
  icmp-ageout
  interface-flow-limit
  flow statistics enable
  no ...
  tcp-ageout
  track icmp
  udp-ageout
```

Description

Configures flow tracking for TCP and UDP flows, and optionally, ICMP flows. The **no** form of this command deletes the flow tracking configuration context.

In order to optimize the flow removal process, flows that have aged-out are flushed in batches. A flow that has aged out is flushed only when the next batch processes. This can cause some flows to stay inactive for a slightly longer time than the value configured here.

Parameter	Description
`enable`	Enables flow tracking.
`icmp-ageout`	Configures an age-out time for ICMP flows, in seconds. Range: 10-86400. Default: 15.

Parameter	Description
`interface-flow-limit`	Configures global concurrent flow limit for flow tracking enabled interfaces. Range: 64-5000. Default: none.
`flow statistics`	Enable flow statistics collection.
`tcp-ageout`	Configures age-out time for established TCP flows in seconds. Range: 120-86400. Default: 600.
`track icmp`	Enable tracking of ICMP flows, in addition to the TCP/UDP flows tracked by default.
`udp-ageout`	Configures age-out time for established UDP flows in seconds. Range: 30-86400. Default: 30.

Examples

Configuring flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)#
```

Deleting flow tracking:

```
switch(config)# no flow-tracking
switch(config)#
```

Enabling flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# enable
```

Disabling flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no enable
```

Configuring an established ICMP flow age-out to 600 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# icmp-ageout 600
```

Removing an established ICMP flow age-out of 600 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no icmp-ageout 600
```

Configuring an established TCP flow age-out to 1000 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# tcp-ageout 1000
```

Removing an established TCP flow age-out of 1000 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no tcp-ageout 1000
```

Configuring an established UDP flow age-out to 1000 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# udp-ageout 1000
```

Removing an established UDP flow age-out of 1000 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no udp-ageout 1000
```

Configuring global level interface flow limit to 256 interfaces:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# interface-flow-limit 256
```

Removing global level interface flow limit to 256 interfaces:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no interface-flow-limit 256
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced on 6200 Switch Series

Command Information

Platforms	Command context	Authority
6200	`config`	Administrators or local user group members with execution rights for this command.

ip|ipv6 flow monitor (interface)

`[no] ip|ipv6 flow monitor (interface)`

Description

Enable flow monitoring on inbound and outbound interfaces by assigning a flow monitor to that interface. Only physical interfaces and LAG interfaces can be monitored. A flow monitor cannot be applied to an interface that is part of a LAG. If an unsupported application is attempted, an error message will be displayed. If the flow monitor is associated with a flow record that contains application fields as collect fields, then Application Recognition should be enabled on the same interface.

The **[no]** form of command disables the flow monitoring.

Examples

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced. on 9300S and 6200 Switch Series

Command Information

Platforms	Command context	Authority
6200	`config` `config-flow-monitor`	Administrators or local user group members with execution rights for this command.

ipv4|ipv6 flow monitor (role)

`[no] ip|ipv6 flow monitor <name>`

Description

Enable flow monitoring on a role. The authorization status of a client does not depend on the flow monitor status for the client in hardware. The client will be authorized even if the system is not able to apply the flow monitor configuration in the role.

In case of multiple clients onboard to a port with varied flow monitor configurations, the flow monitor associated with the first authenticated client on the port will be applied for all the traffic on the port.

The **[no]** form of command removes the flow monitor from the role.

Parameter	Description
`<name>`	Name of the flow monitor , up to 64 characters.

Examples

Enable an IPv4 flow monitor on a role named role01

```
switch# config terminal
switch(config)# port-access role role01
switch(config-pa-role)# ip flow monitor flow-monitor-1
```

Enable an IPv6 flow monitor on a role named role01

```
switch# config terminal
switch(config)# port-access role role01
switch(config-pa-role)# ipv6 flow monitor flow-monitor-2
```

Command History

Release	Modification
10.15	Command introduced on 6200 Switch Series

Command Information

Platforms	Command context	Authority
6200	`config` `config-pa-role`	Administrators or local user group members with execution rights for this command.

show flow monitor

>

Description

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: monitor does not exist)
- Rejected (A record must be assigned to the monitor, but no record is assigned)
- Rejected (The state of the assigned record is rejected)
- Rejected (Internal error: failure in processing the record configuration)
- Rejected (The state of one or more of the assigned flow exporters is rejected)

Parameter	Description
`<name>`	Name of the flow monitor.
`statistics`	Display additional flow and cache statistics.

The possible statistics for a flow monitor are:

Examples

Display the configuration of a flow monitor named **flow-monitor-1**.

```
switch# show flow monitor monitor-1
--------------------------------------------------------------------------------
Flow monitor 'monitor-1'
--------------------------------------------------------------------------------
Description              : Used for IPv4 traffic analysis
Status                   : Accepted
Flow Record              : record-1
Flow Exporter(s)         : exporter-1
```

Display the statistics of a flow monitor named **flow-monitor-1**.

```
switch# show flow monitor monitor-1 statistics
--------------------------------------------------------------------------------
Flow monitor 'monitor-1'
--------------------------------------------------------------------------------
Current Entries          : 2
Flows Added              : 4
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Display information for all flow monitors

```
switch# show flow monitor
--------------------------------------------------------------------------------
Flow monitor 'monitor-1'
--------------------------------------------------------------------------------
Description              : Used for IPv4 traffic analysis
Status                   : Accepted
Flow Record              : record-1
Flow Exporter(s)         : exporter-1, exporter-2


Flow Collector           : collector-1
Flow Exporter(s)         : exporter-1


Display information with no flow monitors configured
```
switch# show flow monitor
No flow monitors configured.
```

```
Display statistics for all flow monitors
```

```
switch# show flow monitor statistics
--------------------------------------------------------------------------------
Flow monitor 'monitor-1'
--------------------------------------------------------------------------------
Current Entries           : 2
Flows Added               : 6
Total Flows Terminated    : 4
  Flows Aged              : 2
    Active Timeout        : 1
    Inactive Timeout      : 1

Total Flows Terminated    : 2
  Flows Aged              : 2
    Inactive Timeout      : 2

  End of Flow Detected    : 2
  Forced End              : 0
```

```

--------------------------------------------------------------------------------
```
```

```
Flow monitor 'monitor-2'
--------------------------------------------------------------------------------
Current Entries            : 6
Flows Added                : 12
```

Command History

Release	Modification
10.15	Command introduced on 9300S and 6200 Switch series

Command Information

Platforms	Command context	Authority
6200	`config` `config-flow-monitor`	Administrators or local user group members with execution rights for this command.

show flow record

```
show flow record [<name>]
```

Description

Display flow record configuration and status. When no record name is specified, the output of this command displays information for all flow records.

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: failed to process record)
- Rejected (Mix of IPv4 and IPv6 match fields is not allowed. Specify match fields of the same IP version (IPv4 or IPv6))

Parameter	Description
`<name>`	Name of the flow record.

Examples

IPv6 related commands are only applicable to switches that support IPv6 protocol.

Display the configuration of a flow record named **flow-record-1**.

```
switch# show flow record record-1
--------------------------------------------------------------------------------
Flow record  'record-1'
--------------------------------------------------------------------------------
Description               : Used for IPv4 traffic analysis
Status                    : Accepted
Match Fields
    ipv4 destination address
```

```
    ipv4 protocol
    ipv4 source address

    transport destination port
    transport source port
Collect Fields
    application name
    counter bytes
    counter packets
    application https URL
```

Display the information of a specific flow record.

```
switch# show flow record record-1
--------------------------------------------------------------------------------
Flow record  'record-1'
--------------------------------------------------------------------------------
Description             : Used for IPv4 traffic analysis
Status                  : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    ipv4 version
    transport destination port
    transport source port
Collect Fields
    application https url
    counter bytes
    counter packets
```

Display information for all flow records

```
switch# show flow record
--------------------------------------------------------------------------------
Flow record  'record-1'
--------------------------------------------------------------------------------
Description             : Used for IPv4 traffic analysis
Status                  : Accepted
Match Fields
        ipv4 destination address
        ipv4 protocol
        ipv4 source address
        ipv4 version
        transport destination port
        transport source port
Collect Fields
        counter bytes
        counter packets
--------------------------------------------------------------------------------
Flow record  'record-2'
--------------------------------------------------------------------------------
Description             : Used for IPv6 traffic analysis
Status                  : Accepted
Match Fields
        ipv6 destination address
        ipv6 protocol
        ipv6 source address
        ipv6 version
```

```
        transport destination port
        transport source port
Collect Fields
        application name
        counter bytes
        counter packets
```
```
Display information for a specific flow record

```
switch# show flow record record-3
--------------------------------------------------------------------------------
Flow record  'record-3'
--------------------------------------------------------------------------------
Description             : Used for IPv4 traffic analysis
destination address, protocol, transport destination port, and transport source
port.)
Match Fields
ipv4 destination address
ipv4 protocol
ipv4 source address
Collect Fields
counter bytes
counter packets
```

Display information with no flow records configured

```
switch# show flow record
No flow records configured
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced on 6200 and 9300S Switch series

Command Information

Platforms	Command context	Authority
6200	Manager (#)	Administrators or local user group members with execution rights for this command.

show flow-tracking

```
show flow-tracking
```

Description

Displays flow-tracking and statistics collection configurations and status.
```

Examples

Display the configuration of role based flow tracking on a 6200 Switch series.

```
switch (config)# show flow-tracking
Flow Tracking Global Configuration
Configuration status            : Enabled
Operational status              : Enabled
Failure Reason                  : NA
UDP Ageout                      : 60  (Seconds)
TCP Ageout                      : 600 (Seconds)
ICMP Ageout                     : 15  (Seconds)
Interface Flow limit            : None
Tracked Protocols               : TCP, UDP
Statistics Collection
Configuration Status         : Enabled
Operational Status           : Enabled
Failure Reason               : NA
Flow Tracking Port Configuration
Interface         App Recognition      IPFIX         Operation Status
-----------         -------------      ----------       ----------
1/1/1               Disabled           Disabled         Disabled
1/1/2               Disabled           Disabled         Disabled
1/1/3               Disabled           Disabled         Disabled
1/1/4               Disabled           Disabled         Disabled
1/1/5               Disabled           Disabled         Disabled
1/1/6               Disabled           Disabled         Disabled
1/1/7               Disabled           Disabled         Disabled
1/1/8               Disabled           Disabled         Disabled
1/1/9               Disabled           Disabled         Disabled
1/1/10              Disabled           Disabled         Disabled
1/1/11              Disabled           Disabled         Disabled
1/1/12              Disabled           Disabled         Disabled
1/1/13              Disabled           Disabled         Disabled
1/1/14              Disabled           Disabled         Disabled
1/1/15              Enabled            Disabled         Enabled
1/1/16              Disabled           Disabled         Disabled
1/1/17              Disabled           Disabled         Disabled
1/1/18              Disabled           Disabled         Disabled
1/1/19              Disabled           Disabled         Disabled
1/1/20              Disabled           Disabled         Disabled
1/1/21              Disabled           Disabled         Disabled
1/1/22              Disabled           Disabled         Disabled
1/1/23              Disabled           Disabled         Disabled
1/1/24              Disabled           Disabled         Disabled
1/1/25              Disabled           Disabled         Disabled
1/1/26              Disabled           Disabled         Disabled
1/1/27              Disabled           Disabled         Disabled
1/1/28              Disabled           Disabled         Disabled
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced on 6200 Switch Series.

Command Information

Platforms	Command context	Authority
6200	Manager (#)	Administrators or local user group members with execution rights for this command.

show tech ipfix

`show tech ipfix`

Description

Shows the IPFIX configuration settings.

If applicable source IP address or source interface is configured for the IPFIX protocol, that configuration is used.

For 6200, 6300,6400,8360,8100 and 9300S Switch Series, If a valid source is configured, the exporter sends flows to an external collector using the effective configured source IP address as the source IP address of the flow packets. In the context of this application, a valid source IP address is any IP address configured in the exporter's VRF namespace.

Examples

The example shows the IPFIX configuration settings.

```
switch#show tech ipfix
=====================================================
Show Tech executed on Tue Apr 11 02:43:06 2023
=====================================================
=====================================================
[Begin] Feature ipfix
=====================================================
*********************************
Command : show flow exporter
*********************************
--------------------------------------------------------------------------------
Flow exporter 'ipfix'
--------------------------------------------------------------------------------
Status                   : Accepted
Export Protocol          : ipfix
Destination Type         : Traffic Insight
Destination              : t1
Transport Configuration
Protocol             : udp
Port                 : 4739
--------------------------------------------------------------------------------
Flow exporter 'V6E1'
--------------------------------------------------------------------------------

....

=====================================================
[End] Feature ipfix
=====================================================
```

For more information on features that use this command, refer to the Security Guide for your switch model.

Command History

Release	Modification
10.15	Command introduced. on 9300S and 6200 Switch Series

Command Information

Platforms	Command context	Authority
6200	Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 4

Boot commands

boot set-default

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
`primary`	Selects the primary network operating system image.
`secondary`	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
`primary`	Selects the primary operating system image for this reboot and sets the configured default operating system image to **primary** for future reboots.
`secondary`	Selects the secondary operating system image for this reboot and sets the configured default operating system image to **secondary** for future reboots.
`serviceos`	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the **boot set-default** command to change the configured default operating system image.

- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

.

To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
`all`	Optional. Shows boot information for the active management module.
`vsf member <1-10>`	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
`Index`	The position of the boot in the history file. Range: 0 to 3.
`Boot ID`	A unique ID for the boot . A system-generated 128-bit string.
`Current Boot, up for <time>`	For the current boot, the **show boot-history** command shows the number of seconds the module has been running on the current software.
`<Timestamp>: boot reason`	For previous boot operations, the **show boot-history** command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: ■ ***<DAEMON-NAME>* crash**: The daemon identified by *<DAEMON-NAME>* caused the module to boot. ■ **Kernel crash**: The operating system software associated with the module caused the module to boot. ■ **Uncontrolled reboot**: The reason for the reboot is not known. ■ **Reboot requested through database**: The reboot occurred because of a request made through the CLI or other API.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================

Index : 2
```

```
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs

Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2

Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1

Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 :  Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 :  Switch boot count is 3
25 Aug 23 10:29:02 :  Primary overtemperature fault detected with 0x2 in PSU 1/1
```

Showing the boot history of the active management module and all line modules:

```
switch#
Management module
=================

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

Management module
=================

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
```

```
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
```

The following example displays the boot history for the VSF member **2**.

```
switch# show boot-history vsf member 2

Member-2
=========

Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs

Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2
```

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 5
External storage

The switch has limited capacity to store data, collected by switch features and protocols. You can provide virtually unlimited storage capacity by adding user-supplied external storage volumes. Supported volume types and storage protocols include: NFSv3, NFSv4, and SCP (sshfs).

One application of external storage is the saving and restoring of DHCP lease files over SCP or NFS network attached storage systems. SCP file system protocol uses a user mode process to emulate a network file system. The key advantage is packet level encryption and simple configuration. The key disadvantage is slow performance.

You can set up external storage volume credentials and then enable it. A storage management process acts on your requests by enabling the storage volume using the requested storage protocol. You can disable the external storage volume or set it up but leave it disable.

The feature maintains storage volume state. The states are: *disabled* (down), *connecting* (establishing connection), *operational* (up), and *unaccessible* (unavailable).

If a storage volume is unavailable, the system attempts to reconnect periodically. Multiple volumes could connect concurrently. If one connection times out the others can connect immediately.

The system supports server connection through data and management ports.

Data port support requires server IP address on a default VRF.

Once a storage volume is enabled, applications can use the volume to store retrieve and delete files and directories.

External storage commands

address (external storage)

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

Description

Specifies the NAS IP address or hostname.

The **no** form of this command deletes an IP address or hostname.

Parameter	Description
<IPV4-ADDR>	Specifies the NAS server IPv4 address, Global.
<IPV6-ADDR>	Specifies the IPv6 address of the NAS server.
<HOSTNAME>	Specifies the hostname of the NAS server. String.

Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no address 10.1.1.1
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution rights for this command.

directory

```
directory <DIRECTORY-NAME>
no directory <DIRECTORY-NAME>
```

Description

Selects an existing directory on the external storage volume.

The **no** form of this command clears a directory of an external storage volume.

Parameter	Description
<DIRECTORY-NAME>	Specifies the external storage directory for mapping the volume.

Examples

Creating a volume named logfiles that is mapped under /home on the server:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory /home:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no directory /home
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

disable external-storage logfiles

```
disable
no disable
```

Description

Disables the external storage volume.

The **no** form of this command enables the external storage volume. This is identical to the `enable` command.

Examples

Disabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Operators or Administrators or local user group members with execution

Platforms	Command context	Authority
		rights for this command. Operators can execute this command from the operator context (>) only.

enable (external-storage logfiles)

```
enable
no enable
```

Description

Enables the external storage volume.

The **no** form of this command disables the external storage volume. This is identical to the `disable` command.

Examples

Creating and then enabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# enable
```

Disables the external storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

external-storage

```
external-storage <VOLUME-NAME>
no external-storage <VOLUME-NAME>
```

Description

Creates or updates an external storage volume.

The **no** form of this command deletes an external storage volume.

Examples

Creating the logfiles storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)#
```

Deleting the logfiles storage volume:

```
switch(config)# no external-storage logfiles
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

password (external-storage)

```
password [{plaintext | ciphertext} <PASSWORD>]
no password {plaintext | ciphertext} <PASSWORD>
```

Description

Sets the password for network attached storage server login.

The **no** form of this command clears the password for network attached storage server login.

Parameter	Description
`{ciphertext	plaintext}`	Selects the password format.
<PASSWORD>	Specifies the password. **NOTE:** When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Creating a volume named logfiles with password Xj#9:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# password plaintext Xj#9
```

Creating a volume named bak1 with a prompted plaintext password:

```
switch(config)# external-storage bak1
switch(config-external-storage-bak1)# password
Enter the NAS server password: **********
Re-Enter the NAS server password: **********
```

Clearing the password for volume logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no password plaintext Xj#9
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution rights for this command.

show external-storage

`show external-storage [<VOLUME-NAME>]`

Description

Shows external storage configuration and state for all volumes or for a specified volume.

Parameter	Description
<VOLUME-NAME>	Specifies the external storage volume name that the show command will use.

Examples

```
switch# show external-storage

---------------------------------------------------------------------------------
--
```

```
            Address       VRF      Username       Type        Directory      State
-----------------------------------------------------------------------------------
--
nfsvol      10.1.1.1      nas      ---            NFSv3       /home
operational
nfsfiles    20.1.1.1      nas      netstorage     NFSv4       /netstor       disabled
scpdev      nasserver     nas      scpstor        SCP         /scp
unaccessible
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config external-storage

`show running-config external-storage`

Description

Shows the running configuration of the external storage.

Examples

```
switch# show running-config external-storage

external-storage nfsvol
      address   10.1.1.1
      vrf       nas
      type      nfsv4
      directoty /home
      enable
external-storage scpdev
      address   30.1.1.1
      vrf       nas
      username  switchuser
      password  ciphertext xxx
      type      scp
      directoty /home
      enable
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

type (external storage)

```
type {nfsv3 | nfsv4 | scp}
no type {nfsv3 | nfsv4 | scp}
```

Description

Sets the network attached storage access type for reaching the external storage volume.

The **no** form of this command deletes an external storage volume.

Parameter	Description
`nfsv3`	Specifies the NFSv3 network access protocol.
`nfsv4`	Specifies the NFSv4 network access protocol.
`scp`	Specifies the SCP network access protocol.

Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no type nfsv4
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution rights for this command.

username (external storage)

```
username <USER-NAME>
no username <USER-NAME>
```

Description

Sets the username for logging in to a network attached storage server.

The **no** form of this command clears a username.

Parameter	Description
<USER-NAME>	Specifies the username.

Examples

Creating a volume named logfiles with the user name nassuser:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name nasuser from accessing the logfiles volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no username nasuser
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution rights for this command.

vrf (external storage)

```
vrf <VRF-NAME>
no vrf <VRF-NAME>
```

Description

Setting a VRF to reach network attached storage.

The **no** form of this command clears access of a VRF to network attached storage.

Parameter	Description
<VRF-NAME>	Specifies the VRF name.

Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no vrf nas
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-external-storage-<VOLUME-NAME>`	Administrators or local user group members with execution rights for this command.

Chapter 6
IP-SLA

The IP Service Level Agreement (IP-SLA) is a feature that enables the measuring of network performance between two nodes in a network for different service level agreement parameters such as round-trip time (RTT), one-way delay, jitter, reachability, packet loss, and voice quality scores. These two nodes can span across area in access, distribution or core inside a LAN as well as across WAN between core to core or core to Data Centre switches. This feature helps you measure the SLA for different protocols or applications such as UDP echo, UDP jitter (for voice and video), TCP connect, HTTP, and ICMP echo. This guide provides details for managing and monitoring different types of IP-SLAs.

IP-SLA guidelines

- AOS-CX supports only SLA configuration through CLI and thresholds can be configured using NAE agents using WebUI/REST.
- AOS-CX supports only forever tests. On-demand tests are not supported.
- Maximum sessions: IP-SLA source 50, IP-SLA responder 8.
- NAE can effectively monitor a maximum of 300 parameters, reducing the maximum supported session by 300.
- NAE supports only syslog.
- NAE agents must be triggered for each IP-SLA test on every switch.
- If multiple IP addresses are received for a DNS query, DNS works with the first resolved IP.
- When the DNS server IP is not configured, the first DNS server in `resolve.conf` is used.
- The source interface/IP option is not applicable for SLAs configured on 'mgmt' VRF, as it has only one interface.
- A system time change because of NTP or a manual change causes an incorrect calculation.
- There is no interoperability of UDP echo SLA between AOS-CX and FlexFabric switches.
- Source IP and source port combination must be unique across SLA sessions in a same switch.
- Do not use the same source port across the source and responder sessions in a switch.
- NTP synchronization is a must for SLA types involving one-way delay such as UDP jitter VoIP.
- It is mandatory to set default CoPP to the max value when UDP jitter SLA is enabled otherwise 100% packet loss can be seen and `UDP-Jitter sla` probe will result in failure as seen in the following example.

```
copp-policy default
     class hypertext priority 6 rate 50000 burst 64
     default-class priority 6 rate 99999 burst 9999
```

- Deviations with respect to PVOS results: The packet losses due to internal switch-related issues like interface shutdown or interface flaps will not be considered as 'Probes Timed-out error', as the IP-SLA solution is to measure network performance and anomalies. Rather, this kind of packet loss will be counted in internal counters like 'Destination address unreachable'.

Limitations with VoIP SLAs

- A maximum of 80 concurrent VoIP SLAs can be scheduled in a 20 second slot.
- A single VoIP probe takes 20 seconds to complete.
- The default and minimum probe interval for VoIP SLA is 120 seconds.
- SLAs scheduled in the same slot, periodically sends 1000 probe packets for 120 seconds in 20 second intervals.
- Default 120 second probe interval is divided in to 6 slots of 20 seconds to avoid synchronization of all configured VoIP SLAs sending probes at the same time.
- SLAs started at the same time exceeding the concurrent limit of 80 must wait for the next 20 second VoIP slot to open before moving to ‘running’ state.
- The maximum number of VoIP SLAs supported is 80 X 6 slots = 480 SLAs.
- SLAs exceeding 480 will continue to remain in the 'waiting for VoIP slot' until any slot is freed by stopping the running SLA.
- To avoid high RTT, a single switch with more than 20 SLAs should not have single responder SLA.
- When IP is received dynamically (e.g. using DHCP) for interfaces other than management interface, IPSLA source or responder has to be configured only using interface name.

IP-SLA commands

https

```
https {get | raw} URL [source {<SOURCE-IPV4-ADDR> | <IFNAME>} source-port <PORT-NUM>]
     [proxy proxy-url] [cache disable] [name-server <IPV4-ADDR-DNS-SERVER>]
     [probe-interval <<PROBE-INTERVAL>>] [version <VERSION-NUMBER>] [https-raw-request
<RAW-PAYLOAD>]
no https {get | raw} URL [source {<SOURCE-IPV4-ADDR> | <IFNAME>} source-port <PORT-NUM>]
     [proxy proxy-url] [cache disable] [name-server <IPV4-ADDR-DNS-SERVER>]
     [probe-interval <<PROBE-INTERVAL>>] [version <VERSION-NUMBER>] [https-raw-request
<RAW-PAYLOAD>]
```

Description

Configures HTTPS as the IP-SLA test mechanism. Requires destination URL and type of HTTPS request (get/raw).

The **no** form of this command removes the configuration.

For HTTPS IP-SLA sessions, it is not required to install a certificate on the switch.

Parameter	Description
{get | raw}	Selects HTTPS request type as get or raw where the system will generate or provide HTTPS payload.
URL	Specifies HTTPS URL address of syntax. https://<HOST NAME/IP-ADDRESS>:<PORT>/<PATH>.
source {<SOURCE-IPV4-ADDR> | <IFNAME>}	Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

Parameter	Description
`source-port <PORT-NUM>`	Specifies the value of the source port for the IP-SLA probes.
`cache disable`	Selects cache option for the HTTPS server. By default the option is enabled.
`name-server <IPV4-ADDR-DNS-SERVER>`	Specifies the IPv4 address of DNS server.
`probe-interval <PROBE-INTERVAL>`	Specifies the probe interval in seconds. Range: 30 to 604800.
`version <VERSION-NUMBER>`	Specifies the source interface to use for sending IP-SLA probes.
`https-raw-request <RAW-PAYLOAD>`	HTTPS raw request. String.

Examples

```
switch(config-ipsla-1)# https get https://device.arubanetworks.com/root/home.html
switch(config-ipsla-1)# https get https://2.2.2.2 source 1/1/1
switch(config-ipsla-1)# https get https://device.arubanetworks.com  source 2.2.2.1
switch(config-ipsla-1)# https get https://device.arubanetworks.com/root/home.html
source-interface 1/1/1
switch(config-ipsla-1)# https get https://device.arubanetworks.com  name-server
10.10.10.2
switch(config-ipsla-1)# https raw https://device.arubanetworks.com/root/home.html
raw-request "GET /en/US/hmpgs/index.html"
switch(config-ipsla-1)# no https get https://2.2.2.2 source 1/1/1
switch(config-ipsla-1)# no https raw
https://device.arubanetworks.com/root/home.html raw-request "GET
/en/US/hmpgs/index.html"
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ip-sla-<IP-SLA-NAME>`	Administrators or local user group members with execution rights for this command.

ip-sla responder

```
ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip) <PORT-NUM>
     {source {< {A.B.C.D | A:B::C:D> | <IFNAME>}|[vrf <VRF-NAME>]{ipv6}
no ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip) <PORT-NUM>
     {source {< {A.B.C.D | A:B::C:D> | <IFNAME>}|[vrf <VRF-NAME>]{ipv6}
```

Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments.

The **no** form of this command removes the IP-SLA responder.

Parameter	Description
<SLA-NAME>	Specifies the SLA name. Length: 1 to 64 characters.
`udp-echo`	Enables responder for udp-echo probes.
`tcp-connect`	Selects TCP connect as the IP-SLA test mechanism.
`vrf` *<VRF-NAME>*	Specifies the name of the VRF to use.
`udp-jitter-voip`	Selects VOIP jitter as the IP-SLA test mechanism.
<PORT-NUM>	Specifies the port number to listen for IP-SLA probes. Range: 1 to 65535.
[source {*<SOURCE-IPV4-ADDR>* | *<IFNAME>*}]	Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.
[source {*<SOURCE-IPV6-ADDR>* | *<IFNAME>*}]	Selects the source IPv6 address for SLA probes or the source interface to use for sending IP-SLA probes.

Usage

The IPv6 keyword is required if an IPv6 address is being used by the source interface or VRF. Otherwise, by default, it will be considered as an IPv4 address.

Examples

```
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 1/1/1
```

```
switch(config)# no ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

```
switch(config)#ip-sla responder <SLA-NAME> udp-jitter-voip 1025 vrf <VRF>
switch(config)#ip-sla responder <SLA-NAME> udp-echo 8000 source 1002::1
switch(config)#ip-sla responder <SLA-NAME> udp-echo 8000 source 1/1/1 ipv6
switch(config)#ip-sla responder <SLA-NAME> udp-jitter-voip 1025 vrf <VRF> ipv6
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Added **ipv6** parameter.

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

show ip-sla responder

`show ip-sla responder <SLA-NAME>`

Description

Shows the given IP-SLA responder configuration and operation status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.

Examples

```
switch(config)# show ip-sla responder SLA3

    SLA Name             : SLA3
    IP-SLA Type          : Udp-echo
    VRF                  : Default
    Responder Port       : 8000
    Responder IP         : 2.2.2.3
    Responder Interface : 1/1/1
    Responder Status     : Running

switch(config)# show ip-sla responder 1

  SLA Name             : 1 (non-persistent)
  SLA Type             : udp-echo
  VRF Name             : default
  Responder Port       : 10
  Responder IP         :
  Responder Interface :
  Responder Status     : Running
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

show ip-sla responder results

`show ip-sla responder <SLA-NAME> <SOURCE-IPV4-ADDR> <PORT-NUM> results`

Description

Shows the given ip-sla responder statistics for a given source IP and port. This command is only applicable for the sources where source IP and port are configured.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
<SOURCE-IPV4-ADDR>	Specifies the source IPV4 address.
<PORT-NUM>	Specifies the port number. Range: 1 to 65535.

Examples

```
switch# show  ip-sla responder SLA1 2.2.2.1 8000 results

    IP-SLA Type          : Udp-echo
    VRF Name             : Default
    Source IP            : 2.2.2.1
    Source Port          : 8000
    Responder Port       : 8888
    Responder IP         : 2.2.2.3
    Responder Interface  :
    Responder Status     : Running
    Packets Received     : 2
    Packets Sent         : 2
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

show ip-sla

```
show ip-sla {<SLA-NAME> [results] | all}
```

Description

Shows the given IP-SLA source configuration and status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
`results`	Shows the statistics calculated for an SLA type.
`all`	Shows all ip-sla source configurations and status.

Examples

```
switch# show ip-sla xyz results

    IP-SLA session status
       IP-SLA Name                       : xyz
       IP-SLA Type                       : tcp-connect
       Destination Host Name/IP Address: 2.2.2.1
       Destination Port                  : 8888
       Source IP Address/IFName          : 2.2.2.2
       Source Port                       : 5555
       Status                            : running

    IP-SLA session cumulative counters
       Total Probes Transmitted          : 1
       Probes Timed-out                  : 0
       Bind Error                        : 0
       Destination Address Unreachable : 0
       DNS Resolution Failures           : 0
       Reception Error                   : 0
       Transmission Error                : 0

    IP-SLA Latest Probe Results
       Last Probe Time                   : 2018 Jul 13 02:00:35
       Packets Sent                      : 1
       Packets Received                  : 1
       Packet Loss in Test               : 0.0000%

    Minimum RTT(ms)                      : 12
    Maximum RTT(ms)                      : 12
    Average RTT(ms)                      : 12
    DNS RTT(ms)                          : 0
    TCP RTT(ms)                          : 12

switch(config)# show ip-sla xyz
    IP-SLA Name                : xyz
    Status                     : scheduled
    IP-SLA Type                : tcp-connect
    VRF                        : ipslasrc
    Source Port                : 5555
    Source IP                  : 2.2.2.2
    Source Interface           :
    Domain Name Server         :
    Probe interval(seconds) : 90


switch(config)# show ip-sla jitter-sla results
```

```
    IP-SLA session status
       IP-SLA Name                       : jitter-sla
       IP-SLA Type                       : udp-jitter-voip
       Destination Host Name/IP Address: 2.2.2.1
       Destination Port                  : 8888
       Source IP Address/IFName          :
       Source Port                       : 5555
       Status                            : running

    IP-SLA Session Cumulative Counters
       Total Probes Transmitted          : 1
       Probes Timed-out                  : 0
       Bind Error                        : 0
       Destination Address Unreachable : 0
       DNS Resolution Failures           : 0
       Reception Error                   : 0
       Transmission Error                : 0

    IP-SLA Latest Probe Results
       Last Probe Time                   : 2018 Jul 13 02:02:48
       Packets Sent                      : 1
       Packets Received                  : 1
       Packet Loss in Test               : 0.0000%

       Minimum RTT(ms)                   : 1
       Maximum RTT(ms)                   : 1
       Average RTT(ms)                   : 1
       DNS RTT(ms)                       : 0

       Min Positive SD                   : 1       Min Positive DS         : 2
       Max Positive SD                   : 1       Max Positive DS         : 2
       Positive SD Number                : 2       Positive DS Number      : 2
       Positive SD Sum                   : 2       Positive DS Sum         : 4
       Positive SD Average               : 5       Positive DS Average     : 5
       Min Negative SD                   : 1       Min Negative DS         : 1
       Max Negative SD                   : 1       Max Negative DS         : 1
       Negative SD Number                : 2       Negative DS Number      : 4
       Negative SD Sum                   : 2       Negative DS Sum         : 4
       Negative SD Average               : 5       Negative DS Average     : 5

       Max SD Delay                      : 0       Max DS Delay            : 0
       Min SD Delay                      : 0       Min DS Delay            : 0
       Average SD Delay                  : 0       Average DS Delay        : 0

    Voice Scores:
       MOS  Score                        : 4.38    ICPIF                   : 0


switch(config)# show ip-sla m3op
    IP-SLA Name              : jitter-sla
    Status                   : running
    IP-SLA Type              : udp-jitter-voip
    VRF                      : ipslasrc
    Source IP                : 2.2.2.2
    Source Interface         :
    Domain Name Server       :
    TOS                      : 10
    Probe Interval(seconds)  : 90
    Advantage Factor         : 0
    Codec Type               : g711a
```

```
switch(config)# show ip-sla https-sla
    SLA Name                : https-sla
    Status                  : running
    SLA Type                : https
    VRF                     : default
    Source Port             : 1027
    Source IP               : 1.1.1.1
    Source Interface        :
    Domain Name Server      :
    Probe Interval(seconds) : 60
    HTTPS Request Type      : raw
    HTTPS URL               : https://1.1.1.2
    Cache                   : Enabled
    HTTPS Proxy URL         :
    HTTP Version Number     :

switch(config)# show ip-sla all

IP-SLA session status
IP-SLA Name                           : 707 (non-persistent)
IP-SLA Type                           : https
Destination Host Name/IP Address      : NA
Destination Port                      : NA
Source IP Address/IFName              :
Source Port                           :
Status                                : running



IP-SLA Session Cumulative Counters
Total Probes Transmitted              : 1
Probes Timed-out                      : 0
Bind Error                            : 0
Destination Address Unreachable       : 0
DNS Resolution Failures               : 0
Reception Error                       : 0
Transmission Error                    : 0



IP-SLA Latest Probe Results
Last Probe Time                       : 2023 Jun 05 13:10:19
Packets Sent                          : 1
Packets Received                      : 1
Packet Loss in Test                   : 0.0000%



Minimum RTT(ms)                       : 20
Maximum RTT(ms)                       : 20
Average RTT(ms)                       : 20
DNS RTT(ms)                           : 0
TCP RTT(ms)                           : 12
TLS RTT(ms)                           : 8

switch(config)# show ip-sla http-sla
    IP-SLA Name             : http-sla
    Status                  : running
    IP-SLA Type             : http
    VRF                     : ipslasrc
    Source IP               : 2.2.2.2
    Source Interface        :
```

```
    Domain Name Server       : 10.10.10.2
    Probe Interval(seconds)  : 90
    HTTP Request Type        : get
    HTTP/HTTPS URL           : abcd.com/ws/home
    Cache                    : Enabled
    HTTP Proxy URL           :
    HTTP Version Number      : 1.1
    ```

IP-SLA status description
    ```
    | Status                  | Description                                   |
    |-------------------------|-----------------------------------------------|
    | running                 | SLA is fully operational                      |
    | Bind Error              | Another service is using the same source port  |
    | Interface Down          | Interface status is not up
    | Dns Resolution Error    | Failed to resolve destination hostname        |
    | No Route                | No available route to the responder           |
    | Internal Error          | Unexpected error prevents SLA session         |
    | Disabled                | SLA is disabled                               |
    |Configuration Incomplete | Configuration is not complete to enable the SLA|
    ```
IP SLA session cumulative counters description
    ```
    | Status                          | Description
                             |
    |---------------------------------|--------------------------------------------
------------------------------|
    |Probes Timed-out                 | Total numbers of probes failed to receive
response.                       |
    |Bind Error                       | Total numbers of probes transmission failed
as source port not available.|
    |Destination Address Unreachable | Total numbers of probes transmission failed
due to route unavailable.    |
    |DNS Resolution Failures          | Total numbers of probes failed due to DNS
resolution failure.             |
    |Reception Error                  | Total numbers of probes failed due to
internal error in reception.        |
    |Transmission Error               | Total numbers of probes failed due to
internal errr in transmission.      |
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.12.1000	Updated to display **https** as an IP-SLA type.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 7
L1-100Mbps downshift

The speed downshift feature allows the user to link-up at sub-optimal speeds when failing to link-up at the highest advertised speed. There are fixed number of link attempts made to establish link at highest advertised speed and when all of them fail and attempt is made to link-up at a lower possible speed.

This feature requires underlying PHY to have support for the same and hence capability is only added to select set of ports. If a link cannot be established at the highest common denominator within a set number of link attempts, the PHY advertises the next highest speed using auto-negotiation.

Limitations with speed downshift

- Link up may be delayed as certain number of retries are done to establish the link at highest advertise speeds by both link partners before downshifting.
- Link may be established at sub-optimal speed.

L1-100Mbps downshift commands

downshift enable

```
downshift-enable
no downshift-enable
```

Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

Examples

```
switch(config-if)# interface 1/1/1
switch(config-if)# downshift-enable

Warning: this is a non-standard mode for use only when standards-based
auto-negotiation is not able to establish a stable link. Enabling this
may cause the port to link at a lower than expected speed and should
not be used on ports that are members of a LAG. Support calls may require
this feature to be disabled

Continue (y/n)?

switch(config-if)#
```

When automatic downshift is enabled:

```
switch(config-if)# show running-config interface
interface 1/1/1
    downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if)# interface 1/1/1
switch(config-if)# no downshift-enable
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

show interface

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-readable]]
show interface [<IFNNAME>] monitor [human-readable]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
`brief`	Shows brief info in tabular format.
`physical`	Shows the physical connection info in tabular format.
`extended`	Shows additional statistics, including the **tx filtered** and **rx filtered** counters. ■ Rx filter packets are protocol packets received when the

Parameter	Description
	protocol is disabled on the switch and there is only one port in the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LLDP. ■ An example of a Tx filtered packet would be a multicast packet being filtered from going out of the ingress port.
`human-readable`	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G. This is available only in the CLI interface output.
`non-zero`	Shows only non zero statistics.
`LAG`	Shows LAG interface information.
`monitor`	Continuously monitor interface statistics.
`LOOPBACK`	Shows loopback interface information.
`TUNNEL`	Shows tunnel interface information.
`VLAN`	Shows VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0-255
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1-255
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094
`VXLAN`	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1

Examples

Showing information when interface 1/1/1 is configured:

```
MDI mode: MDIX
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: all
Rate collection interval: 300 seconds
Rates                           RX                    TX          Total (RX+TX)
------------- -------------------- -------------------- --------------------
Mbits / sec                   0.00                  0.00                  0.00
KPkts / sec                   0.00                  0.00                  0.00
Unicast                       0.00                  0.00                  0.00
Multicast                     0.00                  0.00                  0.00
Broadcast                     0.00                  0.00                  0.00
Utilization %                 0.00                  0.00                  0.00
Statistics                      RX                    TX                 Total
------------- -------------------- -------------------- --------------------
Packets                          0                     0                     0
Unicast                          0                     0                     0
```

```
Multicast                         0                    0                    0
Broadcast                         0                    0                    0
Bytes                             0                    0                    0
Jumbos                            0                    0                    0
Dropped                           0                    0                    0
Filtered                          0                    0                    0
Pause Frames                      0                    0                    0
Errors                            0                    0                    0
CRC/FCS                           0                  n/a                    0
Collision                       n/a                    0                    0
Runts                             0                  n/a                    0
Giants                            0                  n/a                    0
```

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

Showing information when the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
```

Showing information when the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if)# show interface 1/1/1 physical
---------------------------------------------------------------------------------
----------------------------------------------------------
                            Link     Admin         Speed            Flow-Control
     EEE        PoE Power                       Port
Port         Type          Status    Config    Status | Config     Status | Config
Status | Config   (Watts)     State Information   Description
---------------------------------------------------------------------------------
----------------------------------------------------------
1/1/1        1GbT          up        up        1G       auto       off       off
on       on        --        10M/100M/1G        --
```

Showing the monitor information:

> In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

```
Interface 1/1/1 is up
Rate                                RX                   TX         Total (RX+TX)
---------------- -------------------- -------------------- --------------------
MBits / sec                   30196.43             30196.43             60392.85
MPkts / sec                   58977.39             58977.40            117954.79
Unicast                           0.00                 0.00                 0.00
Multicast                     58977.39             58977.40            117954.79
```

```
Broadcast                           0.00                 0.00                 0.00
Utilization %                      75.49                75.49               150.98
Statistic                             RX                   TX        Total (RX+TX)
---------------- -------------------- -------------------- --------------------
Packets                       4756527649           4756527865           9513055514
Unicast                                0                    0                    0
Multicast                     4756527649           4756527865           9513055514
Broadcast                              2                    0                    2
Bytes                       304417778668         304417795428         608835574096
Jumbos                                 0                    0                    0
Dropped                                0          19028847730          19028847730
Pause Frames                           0                    0                    0
Errors                                 0                    0                    0
CRC/FCS                                0                  n/a                    0
help: ?, quit: q
```

```
Help for Interface Monitor
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q
```

Showing the output for interface 1/1/1 in human-readable format:

In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```
switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up
Rate                                  RX                   TX        Total (RX+TX)
---------------- -------------------- -------------------- --------------------
Bits / sec                            3M                   3M                   6M
Pkts / sec                           316                  316                  633
Unicast                              319                  319                  638
Multicast                              0                    0                    0
Broadcast                              0                    0                    0
Utilization %                        < 1                  < 1                  < 1
Statistic                             RX                   TX                Total
---------------- -------------------- -------------------- --------------------
Packets                             577K                 577K                   1M
Unicast                             577K                 577K                   1M
Multicast                              0                   51                   51
Broadcast                              0                   15                   15
Bytes                               744M                 745M                   1G
Jumbos                                 0                    0                    0
Dropped                                0                    0                    0
Filtered                               0                    0                    0
Pause Frames                           0                    0                    0
Errors                                 0                    0                    0
CRC/FCS                                0                  n/a                    0
Collision                            n/a                    0                    0
Runts                                  0                  n/a                    0
```

```
Giants                                    0                  n/a                    0
```

Showing information about extended counters:

The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
--------------------------------------------------------------------
Interface 1/1/17
--------------------------------------------------------------------
Statistics                                              Value
--------------------------------------------------------------------
Dot1d Tp Port In Frames                                 547
Dot1d Tp Port Out Frames                                608
Dot3 In Pause Frames                                    0
Dot3 Out Pause Frames                                   0
Ethernet Stats Broadcast Packets                        19
Ethernet Stats Bytes                                    40162
Ethernet Stats Packets                                  342
...
--------------------------------------------------------------------
Error-Statistics                                        Value
--------------------------------------------------------------------
Dot1d Base Port MTU Exceeded Discards                   0
Dot3 Control In Unknown Opcodes                         0
Dot3 Stats Alignment Errors                             0
Dot3 Stats FCS Errors                                   0
Dot3 Stats Frame Too Longs                              0
Dot3 Stats Internal Mac Transmit Errors                 0
Ethernet RX Oversize Packets                            0
...
```

Showing interface link-status:

```
switch# show interface link-status
--------------------------------------------------------------


Port           Type           Physical    Link         Last
                              Link State  Transitions  Change
--------------------------------------------------------------
1/1/1          1G-BT          down        0            --
1/1/2          1G-BT          up          1            1 minute ago (Fri Mar 09
12:36:56 UTC 2018)
1/1/3          1G-BT          up          1            1 minute ago (Fri Mar 09
12:36:56 UTC 2018)
1/1/4          --             down        0            --
1/1/5          --             down        0            --
```

Showing interface loopback 1 link-status:

```
--------------------------------------------------------------
                              Physical    Link         Last
Port           Type           Link State  Transitions  Change
```

```
--------------------------------------------------------------
loopback1      --             up          --           --
```

Showing interface 1/1/2-1/1/3 link-status:

```
--------------------------------------------------------------
                              Physical    Link         Last
Port           Type           Link State  Transitions  Change
--------------------------------------------------------------
1/1/2          1G-BT          up          1            1 minute ago (Fri Mar 09
12:36:56 UTC 2018)
1/1/3          1G-BT          up          1            1 minute ago (Fri Mar 09
12:36:56 UTC 2018)
```

Showing interface link-status:

```
switch# show interface link-status
--------------------------------------------------------------------------
Port           Type            Physical    Link        Link Flaps  Last
                               Link State  Transitions Ignored     Change
--------------------------------------------------------------------------
1/1/1          1G-BT           down        0           0           --
1/1/2          1G-BT           up          1           0           1 minute ago
(Fri Mar 09 12:36:56 UTC 2018)
1/1/3          1G-BT           up          1           0           1 minute ago
(Fri Mar 09 12:36:56 UTC 2018)
1/1/4          --              down        0           0           --
1/1/5          --              down        0           0           --
```

Showing state information when interface is blocked:

```
8360(config-if)# show interface 1/1/1

Interface 1/1/1 is up (Blocked)
Admin state is up
State information: Blocked by UDLD
Link state: up for 1 minute (since Mon Jun 10 09:25:27 UTC 2024)
Link transitions: 1
Description:
Persona:
Hardware: Ethernet, MAC Address: 00:fd:45:67:85:91
MTU 1500
Type 10G-LR / 10G SFP+ LR
Full-duplex
qos trust none
Speed 10000 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rate collection interval: 300 seconds

Rate                               RX                   TX        Total (RX+TX)
---------------- -------------------- -------------------- --------------------
Mbits / sec                      0.00                 0.00                 0.00
KPkts / sec                      0.00                 0.00                 0.00
```

```
Unicast                        0.00                 0.00                 0.00
Multicast                      0.00                 0.00                 0.00
Broadcast                      0.00                 0.00                 0.00
Utilization %                  0.00                 0.00                 0.00
Statistic                        RX                   TX                Total
---------------- -------------------- -------------------- --------------------
Packets                          15                   15                   30
Unicast                          12                   12                   24
Multicast                         3                    3                    6
Broadcast                         0                    0                    0
Bytes                          1350                 1350                 2700
Jumbos                            0                    0                    0
Dropped                           0                    0                    0
Pause Frames                      0                    0                    0
Errors                            0                    0                    0
CRC/FCS                           0                  n/a                    0
Collision                       n/a                    0                    0
Runts                             0                  n/a                    0
Giants                            0                  n/a                    0
```

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.15	Added state information when port goes into down state.
10.11	Added `monitor` parameter.
10.10	Added `human-readable` parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface statistics

```
show interface [<IFNAME>|<IFRANGE>] statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] statistics monitor [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics monitor [non-zero] [human-readable]
show interface vxlan <VXLAN-ID> statistics [non-zero] [human-readable]
```

Description

Shows statistics for switch interfaces such as packets transmitted and received, bytes transmitted and received, broadcast and multicast packets.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
`LAG`	Shows LAG interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
`VXLAN`	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1
`monitor`	Continuously monitor interface statistics.
`human-readable`	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G.
`non-zero`	Shows only non zero statistics.

Examples

Showing statistics of all interfaces:

```
show interface statistics
------------------------------------------------------------------------------------------------------------------------------------------------
Interface                 RX Bytes   RX Packets   RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast RX Multicast  TX Broadcast TX Multicast  RX Pause   TX Pa
------------------------------------------------------------------------------------------------------------------------------------------------
1/1/1                      2727136         1975          0     17796         195         0            82         1788            96           54         0
1/1/10                           0            0          0         0           0         0             0            0             0            0         0
1/1/11                           0            0          0         0           0         0             0            0             0            0         0
1/1/12                           0            0          0         0           0         0             0            0             0            0         0
...
1/1/30       - lag1              0            0          0     11271          92         0             0            0             0           51         0
1/1/31       - lag2           2360           25         50   2732119        2040         0             0            0           178         1839         0
1/1/32       - lag2              0            0          0     11373          93         0             0            0             0           51         0
vlan1                            0            0          0         0           0         0             0            0             0            0         0
```

Showing statistics of all interfaces with only non-zero statistics:

```
show interface statistics non-zero
---------------------------------------------------------------------------------------------------------------------------------------------------
Interface                       RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets TX Drops RX Broadcast  RX Multicast  TX Broadcast TX Multicast  RX Pause  TX Pause
---------------------------------------------------------------------------------------------------------------------------------------------------
1/1/1                            2727136        1975         0     17796         195        0            82          1788            96           54         0         0
1/1/30      - lag1                     0           0         0     11271          92        0             0             0             0           51         0         0
1/1/31      - lag2                  2360          25        50   2732119        2040        0             0             0           178         1839         0         0
1/1/32      - lag2                     0           0         0     11373          93        0             0             0             0           51         0         0
```

Showing statistics of all interfaces in the human-readable format:

```
show interface statistics human-readable
--------------------------------------------------------------------------------------------------------------------------------------
Interface             RX Bytes   RX Pkts  RX Drops  TX Bytes   TX Pkts  TX Drops  RX Bcast  RX Mcast  TX Bcast  TX Mcast  RX Pause  TX Pause
--------------------------------------------------------------------------------------------------------------------------------------
1/1/1                     744M      577K         0      745M      578K         0         0         0        73       287         0         0
1/1/2                     474M      367K         0      475M      369K         0         0         0        73       288         0         0
1/1/3                        0         0         0         0         0         0         0         0         0         0         0         0
```

Showing statistics of a single interfaces:

```
show interface 1/1/2 statistics
----------------------------------------------------------------------------------------------------------------------------------------------------
Interface                  RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast  RX Multicast  TX Broadcast  TX Multicast  RX Pause  TX Pause
----------------------------------------------------------------------------------------------------------------------------------------------------
1/1/2                       2725080        1931         0     25877         253         0            21          1788            65            55         0         0
```

Showing statistics of all members of a LAG interface:

```
show interface lag1 statistics
----------------------------------------------------------------------------------------------------------------------------------------------------
Interface                  RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast  RX Multicast  TX Broadcast  TX Multicast  RX Pause  TX Pause
----------------------------------------------------------------------------------------------------------------------------------------------------
1/1/3        - lag1            2424          26         0   2734082        2062         0             0             0           191          1848         0         0
1/1/30       - lag1               0           0         0     12383         100         0             0             0             0            59         0         0
lag1                           2424          26         0   2746465        2162         0             0             0           191          1907         0         0
```

Showing error statistics of all interfaces:

```
show interface error-statistics
--------------------------------------------------------------------------------
Interface                       RX Errors   TX Errors    Giants    Runts   CRC/FCS  Collisions
--------------------------------------------------------------------------------
1/1/1                                 190          20    100647        0         0           0
1/1/10                                  0           0       100      290      7165         949
1/1/11                                  0           0         0        0         0           0
1/1/12                                  0           0         0        0         0           0
...
1/1/30 - lag1                        1500         500     45800        0         0           0
1/1/31 - lag2                           0           0        11       27         0           0
1/1/32 - lag2                           0           0         0        0         6          18
```

Showing monitor statistics:

The rows and columns of show interface monitor statistics depends on the length of width of the client terminal. The CLI can be navigated using the arrow keys as well as the PageUp, PageDown, Home, and End keys.

```
show interface statistics monitor
----------------------------------------------------------------
Interface                        RX Bytes          RX Packets >>
----------------------------------------------------------------
1/1/1                       3440525421984         53758209526
1/1/2                       3440526607008         53758228042
1/1/3                       3440527785312         53758246453
1/1/30                      3440559671264         53758744653
1/1/31                      3440560851680         53758763098
1/1/32                      3440562028704         53758781489

                                                  help: ?, quit: q
```

```
Help for Interface Monitor

f     Toggle full statistics
h     Toggle human-readable mode
n     Toggle non-zero mode
r     Toggle rate display

c     Clear interface statistics
        Does not apply to rates

Arrows, PgUp, PgDn, Home, End
        Navigate interface statistics

Delay:2
                                                                  help: ?, qui
```

Showing monitor error statistics in human-readable format:

```
show interface 1/1/1-1/1/3,1/1/30-1/1/32 error-statistics monitor human-readable
---------------------------------------------------------------------------------------------
Interface                RX Errors    TX Errors    RX Giants     RX Runts      CRC/FCS  Collisions
---------------------------------------------------------------------------------------------
1/1/1                            0            0            0            0            0            0
1/1/2                            0            0            0            0            0            0
1/1/3                            0            0            0            0            0            0
1/1/30                           0            0            0            0            0            0
1/1/31                           0            0            0            0            0            0
1/1/32                           0            0            0            0            0            0


Human-readable                                                                   help: ?, quit: q
```

```
Help for Interface Monitor

h    Toggle human-readable mode
n    Toggle non-zero mode

c    Clear interface statistics
        Does not apply to rates

Arrows, PgUp, PgDn, Home, End
        Navigate interface statistics

Delay:2
                                                                            help: ?, quit: q
```

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.11	Added `moitor` parameter.
10.10	Added `human-readable` parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface downshift-enable

```
show interface [<IFNNAME>|<IFRANGE>] downshift-enable
```

Description

Displays speed downshift information, including the interface speed status and configuration.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.

Examples

Showing automatic downshift information:

```
switch(config-if)# show interface downshift-enable
--------------------------------------------------
              Downshift              Speed
Port      Enabled | Active    Status    | Config
--------------------------------------------------
1/1/1     yes       yes       100M-FDx   auto
1/1/2     yes       no        1G         auto
1/1/3     yes       no        100M-FDx   100M-FDx
1/1/4     no        no        --         auto
```

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
--------------------------------------------------
              Downshift              Speed
Port      Enabled | Active    Status    | Config
--------------------------------------------------
1/1/2     yes       no        1G         auto
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show running-config interface

```
show running-config interface [<IFNNAME>|<IFRANGE>]
show running-config interface [lag | loopback | tunnel | vlan ] [<ID>]
```

Description

Displays active configurations of various switch interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
LAG	Specifies LAG interface information
LOOPBACK	Specifies loopback interface information.
TUNNEL	Specifies tunnel interface information.
VLAN	Specifies VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256.
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0-255.
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1-255.
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094.
VXLAN	Specifies the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1.

Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2

interface 1/1/2
   no shutdown
   description DC-23
   exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback

interface loopback 1
    description lb interface 1
    exit
interface loopback 2
    description lb interface 2
    exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback

No loopback interfaces configured.
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Chapter 8
Mirroring

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.

Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Classifier policies and mirroring sessions

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

1. Configuring a mirroring session with a destination interface.
2. Enabling the mirroring session.
3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

If the packets being mirrored are received from a VLAN that is not allowed on the mirror destination, the mirrored packets would be dropped at the mirror destination interface. When the mirrored packets are dropped at the destination, the mirror output packet and byte count will increment, however the packets will not be received at the mirror destination.

The mirror destination port among the active mirror sessions must be unique. That is, if an interface is configured as a source or destination in an active mirror session, the same port cannot be used as a destination in another active mirror session.

VLAN as a source

AOS-CX allows configuration of VLAN as a mirroring source. When a VLAN source is configured in the 'rx' direction, all packets are mirrored as they are received in the switch. When a VLAN source is configured in 'tx' direction, all packets are mirrored as they are transmitted out of the switch.

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs in each direction of a given mirror session.

Same VLANs can be configured as a mirror source for multiple sessions.

Note: When changing a source VLAN in an enabled mirror session (that is, adding, changing direction, or removing), mirrored packets being transmitted out the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways:

1. Reenter the `source vlan` command with the new preferred direction.
2. Use the `no` form of the command with a direction (rx or tx) to selectively remove the specified direction. Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

For packets bridged through the switch:

If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in the 'rx' direction, packets are mirrored in the pre-routed form with the destination MAC address as the switch address.
- If the mirror is configured in the 'tx' direction, packets are mirrored in the post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in the 'both' direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behaviors for mirroring with VLAN as source:

- If the mirror is configured in the 'rx' or 'tx' direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the 'both' direction, two copies of the packets are mirrored to the mirror destination.

Mirroring commands

clear mirror

`clear mirror [all | <SESSION-ID>]`

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Parameter	Description
`all`	Specifies all configured sessions.
<SESSION-ID>	Specifies a numeric identifier for the session. Range: 1 to 4

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

clear mirror endpoint

`clear mirror endpoint [<NAME>]`

Description

Clears mirror endpoint statistics for all configured mirror endpoints. The optional parameter can be added to clear a specific mirror endpoint.

Parameter	Description
<NAME>	Specifies name of the mirror endpoint instance to be cleared.

Examples

Clearing statistics for all configured mirror endpoints:

```
switch# clear mirror endpoint
```

Clearing mirror statistics for mirror endpoint test:

```
switch# clear mirror endpoint test
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

comment

```
comment <COMMENT>
no comment
```

Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The **no** form of this command removes the comment.

Parameter	Description
<COMMENT>	A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3)# comment This Mirror will be removed during next
maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3)# no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor statistics on each endpoint
interfaces
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>` `config-mirror-endpoint`	Administrators or local user group members with execution rights for this command.

copy tcpdump-pcap

`copy tcpdump-pcap <FILE-NAME> <REMOTE-URL>`

Description

Saves packet capture files to external storage.

Parameter	Description
<FILE-NAME>	Specifies the packet capture file to save.
<REMOTE-URL>	Specifies the external storage to which the packet capture file will be saved.

Usage

Only four files can be saved at any point on the switch. Packet capture files are not saved after a failover or reboot. View a list of saved files using **diag utilities list-files**.

Examples

Saving my_capture_file.pcap to sftp://root@10.0.0.2/file.pcap:

```
switch# copy tcpdump-pcap my_capture_file.pcap sftp://root@10.0.0.2/file.pcap
root@10.0.0.2's passowrd:
Connected to 10.0.0.2.
sftp > put my_capture_file.pcap file.pcap
Uploading my_capture_file.pcap to /root/file.pcap
my_capture_file.pcap                                          100%    156    219.8KB/s    00:00
Copied successfuly.
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

copy tshark-pcap

`copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]`

Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

Parameter	Description
<REMOTE-URL>	Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is: {tftp:// | sftp://**<USER>**@} {**<IP>**|**<HOST>**} [:**<PORT>**] [;blocksize=**<SIZE>**]/<*FILE*>
vrf *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap
```

```
root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
Uploading packets.pcap to /root/file.pcap
packets.pcap                                  100%  156   219.8KB/s   00:00
Copied successfully.
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

destination cpu

```
destination cpu
no destination cpu
```

Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the **Supportability Guide**.

The **no** form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution rights for this command.

destination interface

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}
no destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

Description

Configures the specified interface as the destination of the mirrored traffic.

The **no** form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Parameter	Description
<INTERFACE-ID>	Specifies a interface. Format: `member/slot/port`.
<LAG-NAME>	Specifies a LAG (link aggregation group) identifier.

Usage

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration.

Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Switch	Destination interface limit per mirror session (4 possible sessions)
5420	64
6200	64

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution rights for this command.

destination tunnel

```
destination tunnel <TUNNEL-IPV4> source <SOURCE-IPv4-ADDR>
   dscp <DSCP-VALUE> vrf <VRF-NAME>
no destination tunnel
```

Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror_err_tunnel_oob_port_not_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel_route_resolution_not_populated" operation status.

The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

The **no** form of this command will cease the use of the tunnel and disable the session.

Parameter	Description
<TUNNEL-IPV4-ADDR>	Specifies the tunnel address in IPv4 format (**x.x.x.x**), where **x** is a

Parameter	Description
	decimal number from 0 to 255.
<SOURCE-IPv4-ADDR>	Specifies the source address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.
<DSCP-VALUE>	Specifies the DSCP value to be carried within the DS field of ERSPAN packet header. Range: 0 to 63. Default: 0.
<VRF-NAME>	Specifies a VRF name. Default: default.

Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10 vrf
default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 10 vrf
default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 2 vrf
default
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution rights for this command.

diagnostic

```
diagnostic

diag utilities tshark [file]
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.

The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the **delete-file** form of this command to delete the most recent capture file.

Since **file** and **delete-file** are optional, the behavior of the base command **diag utilities tshark** does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Parameter	Description
`file`	Saves captured packets to a temporary file.
`delete-file`	Deletes the most recent captured file.

Example

Performing diagnostic:

```
switch# diagnostic

switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

diag utilities tcpdump

```
diag utilities tcpdump [command <TEXT> | delete file <FILE-NAME> | list-files |
  vrf <VRF-NAME> | count <COUNT-NUM> | proto <PROTO-NUM> | host-ip <IP-ADDR> | source-ip
  <IP-ADDR> | destination-ip <IP-ADDR> | host-port <PORT> | source-port <PORT> |
  destination-port <PORT> | verbosity <LEVEL> | print <DATA> | ethernet-type <ETH-NUM>]
```

Description

Captures traffic received or transmitted over a network.

Parameter	Description
`command <TEXT>`	Captures packets based on a specified tcpdump command string.
`delete file <FILE-NAME>`	Deletes specified tcpdump list files.
`list-files`	Lists all the tcpdump capture files saved on the device.
`vrf <VRF-NAME>`	Captures packets on the specified VRF. If no VRF is named, the default is used.
`count <COUNT-NUM>`	Runs the tcpdump command until the specified number of packets are captured. Range: 1-2147483647.
`proto <PROTO-NUM>`	Captures packets of a particular type based on IP protocol number. Range: 0-255.
`host-ip <IP-ADDR>`	Captures packets matching with the source or destination IP address.
`source-ip <IP-ADDR>`	Captures packets from the specified IP address.
`destination-ip <IP-ADDR>`	Captures packets sent to the specified IP address.
`host-port <PORT>`	Captures packets matching with the source or destination port.
`source-port <PORT>`	Captures packets from the specified IP port.
`destination-port <PORT>`	Captures packets sent to the specified IP port.
`verbosity <LEVEL>`	Captures packets of the specified verbosity. Range: level1-level4. If no verbosity is specified, the default is level1.
`print <DATA>`	Captures the data of each packet. The maximum is 262144 bytes
`ethernet-type <ETH-NUM>`	Captures packets based on the particular ethernet type. Range: 0-65535.

Usage

- When using the **command** option, the only traffic captured will be packets that have been mirrored to the CPU.
- When using the **command** option, command line sanitization is performed to prevent options that may cause harm or security issues. The following options are blocked:
  - -i/--interface
  - -Z
  - -B/--buffer-size
  - -C

    - -W
    - -Z/--relinquish privileges
- Non-word operators such as "&" or "|" are not allowed. Use boolean keywords such as "and," "or," and "not."
- When using **command -r** to read a file, do not provide any directory path characters. Use list-files command to get the list of file names currently saved on the device, and then use those file names.
- A total of four files can be saved at any given point on the device. Packet capture files are not saved after a failover or reboot, but can be saved to external storage using the **copy tcpdump-pcap** command.

Examples

Inspecting traffic mirrored to the CPU via tcpdump and saving the output to my_capture_file.pcap:

```
switch# diag utilities tcpdump command -c 2 -x -w my_capture_file.pcap
Inspecting traffic mirrored to the CPU via tcpdump until Ctrl-C is entered.
2 packets captured
2 packets received by filter
0 packets dropped by kernel
Ending traffic capture.
```

Listing saved capture files:

```
switch# diag utilities tcpdump list-files
my_capture_file.pcap
```

Reading my_capture_file.pcap:

```
switch# diag utilities tcpdump command -r my_capture_file.pcap
reading from file /tmp/tcpdump/my_capture_file1.pcap, link-type EN10MB (Ethernet)
  1  11:59:34.047867 IP6 localhost.40318 > localhost.ntp: NTPv2, Reserved, length
12
        0x0000:  0000 0304 0006 0000 0000 0000 0000 86dd  ................
        0x0010:  600a 7e47 0014 1140 0000 0000 0000 0000  `.~G...@........
        0x0020:  0000 0000 0000 0001 0000 0000 0000 0000  ................
        0x0030:  0000 0000 0000 0001 9d7e 007b 0014 0027  .........~.{...'
        0x0040:  1601 0001 0000 0000 0000 0000            ............
  2  11:59:34.047915 IP6 localhost.ntp > localhost.40318: NTPv2, Reserved, length
12
        0x0000:  0000 0304 0006 0000 0000 0000 0000 86dd  ................
        0x0010:  6b8d 23c5 0014 1140 0000 0000 0000 0000  k.#....@........
        0x0020:  0000 0000 0000 0001 0000 0000 0000 0000  ................
        0x0030:  0000 0000 0000 0001 007b 9d7e 0014 0027  .........{.~...'
        0x0040:  d681 0001 c016 0000 0000 0000
```

Removing my_capture_file.pcap:

```
switch# diag utilities tcpdump delete-file my_capture_file.pcap
Successfully removed file
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

disable (mirror session)

```
disable
```

Description

Disables the mirroring session specified by the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the **show mirror** command for that session ID shows an **Admin Status** of **disable** and an **Operation Status** of **disabled**.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution rights for this command.

enable (mirror session)

```
enable
```

Description

Enables the mirroring session for the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the **show mirror** command for that session ID shows an **Admin Status** of **enable** and an **Operation Status** of **enabled**.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of **rx** or **both**):

- The attempt to enable the mirroring session fails and an error is returned.

When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution rights for this command.

mirror session

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The **no** form of this command removes an existing mirroring session from the configuration.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#

switch(config)# mirror session 3
switch(config-mirror-3)#

switch(config)# no mirror session 1
switch(config)#
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

mirror endpoint

```
mirror endpoint <NAME>
no mirror endpoint <NAME>
```

Description

Creates the specified mirror endpoint or enters its context if it already exists. The specifics of a mirror endpoint are created or altered while in the mirror endpoint context and the mirror endpoint is enabled or disabled from this context. It may be possible to support different encapsulations by different ASICs. For example, UDP for PVOS compatibility. Termination of GRE encapsulation is also supported.

The **no** form of this command removes an existing mirror endpoint. An enabled mirror endpoint is automatically disabled first before removal.

Parameter	Description
<NAME>	Specifies mirror endpoint name.

Examples

Creating a mirror endpoint named test :

```
switch(config)# mirror endpoint test
```

Deleting mirror endpoint named test:

```
switch(config)# no mirror endpoint test
```

Configuring a mirror endpoint named test :

```
6100(config)# mirror endpoint test
6100(config-mirror-endpoint-test)#
6100(config-mirror-endpoint-test)# destination
  interface  Specify interfaces to send traffic
6100(config-mirror-endpoint-test)# destination interface
  IFNAMELIST  An interface, a range or a comma seperated list of interfaces
6100(config-mirror-endpoint-test)# destination interface 1/1/3
  <cr>
6100(config-mirror-endpoint-test)# destination interface 1/1/3
6100(config-mirror-endpoint-test)#
6100(config-mirror-endpoint-test)# source 1.1.1.1 destination 1.1.1.2 id 1 vrf
default
6100(config-mirror-endpoint-test)#
```

Only physical ports can be configured as interface for mirror-endpoint destination. LAG port is not supported as interface for mirror-endpoint destination.

The maximum allowed number of destination interfaces for both mirror-session and mirror-endpoint is 1.

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.13.1000	Added support for 4100i, 6000, and 6100 switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

show mirror

`show mirror [<SESSION-ID>]`

Description

Shows information about mirroring sessions. If **<SESSION-ID>** is not specified, then the command shows a summary of all configured mirroring sessions. If **<SESSION-ID>** is specified, then the command shows detailed information about the specified mirroring session.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4

Usage

Information in the **Admin Status** column of the command output indicates the configured status. The admin status is one of the following values:

- **enable**: The mirroring session is enabled.
- **disable**: The mirroring session has been configured but not yet enabled, or has been disabled.

Information in the **Operation Statu**s colum indicates the status of the mirroring session. Operation status is one of the following values:

- **dest_doesnt_exis**t:The configured destination interface is not found in the system. The mirroring session cannot be enabled.
- destination_shutdown**: The mirroring session is enabled, but the destination interface is shut down. No traffic** can be monitored.
- **disabled**: The mirroring session is disabled and is not in an error condition.
- **enabled**: The mirroring session is enabled.
- **external/driver_error**: An internal ASIC hardware error occurred.
- **hit_active_sessions_capacity**: The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.
- **internal_error**: An invalid parameter was passed to the ASIC software layer.
- **no_dest_configured**: The mirroring session does not have a destination interface configured.
- **no_name_configured**: A software error occurred. The mirroring session does not have a session ID in its configuration.
- **null_mirror**: A software error occurred. The session object reference is invalid.
- **out_of_memory**: The system is out of memory, reboot recommended.
- **tunnel_route_resolution_not_populated**: If the destination tunnel IP address is not reachable.
- **unknown_error**: An unexpected error occurred.

Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
--- ------------- -----------------------------------------------------
1   enable        enabled
2   disable       disabled
3   disable       disabled
4   enable        internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
 Mirror Session: 3
 Admin Status: disable
 Operation Status: disabled
 Comment: Monitor router port ingress-only traffic
 Source: interface 1/1/2 rx
 Destination: interface 1/1/3
switch#
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mirror endpoint

`show mirror endpoint [<NAME>]`

Description

Shows a list of all configured mirror endpoints, their Admin Status and their Operation Status.

The optional parameter will display the details of the specified mirror endpoint if it exists.

Parameter	Description
<NAME>	Specifies name of the mirror endpoint instance to be displayed.

Examples

Showing a summary of all configured mirror endpoints on the switch:

```
switch# show mirror endpoint
Name    Admin Status   Operation Status
-----  -------------- ----------------------------------------------------
test    enable         enabled
monitor disable        disabled
```

Showing the details of enabled mirror endpoint test:

```
switch# show mirror endpoint test
```

```
Mirror Endpoint: audit
Admin Status: enable
Operation Status: enabled
Comment: Mirror Endpoint Audit
Type: gre
Tunnel: source 1.1.1.1 destination 1.1.1.2 id 1 vrf default
Interface: 1/1/3
Output Packets: 123456789
Output Bytes: 0
```

"Output Packets" in "show mirror endpoint [name]" is only supported for statistics.
"Output Bytes" in "show mirror endpoint [name]" is not supported due to ASIC limitation.

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

shutdown (mirror endpoint)

```
shutdown
no shutdown
```

Description

Enables mirror endpoint from its default disabled state. To verify the mirror endpoint was successfully activated, run the **show mirror endpoint NAME** command and verify that the **Admin Status** and **Operational Status** has changed from disabled to enabled. If the status value remains disabled, consult the system logs to determine the reason for activation failure. To disable the mirror endpoint, first disable the remote mirror session on the switch that's originating the data. Next, use the `shutdown` command to disable the mirror endpoint.

Examples

Enabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# no shutdown
```

Disabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# shutdown
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

source

```
source <SOURCE-IP> destination <DESTINATION-IP> id <1-4294967295> [vrf <VRF_NAME>] [type
{gre}]
no source
```

Description

Configures tunnel parameters of the mirror endpoint. Configuring a tunnel parameter to a mirror endpoint will replace the existing configuration. By default the VRF is **default**, users can also explicitly provide a custom VRF. The default tunnel type is considered to be GRE and users also have the option to explicitly give type as GRE.

The **no** form removes the tunnel parameters of the mirror endpoint.

Parameter	Description
<SOURCE-IP>	Specifies L3 encapsulated IPv4 source in the form A.B.C.D.
<DESTINATION-IP>	Specifies L3 encapsulated IPv4 destination in the form A.B.C.D.
`id`	Specifies tunnel identifier from the encapsulated packet.
<VRF_NAME>	Specifies the name of VRF for which the tunnel belongs to.

Examples

Configuring a tunnel parameter to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# source 1.1.1.1 destination 7.7.7.7 id 1 vrf
default type gre
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

source interface

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored.

The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Parameter	Description
<PORT-NUM>	Specifies a physical port on the switch. Use the format **member/slot/port** (for example, **1/3/1**).
<LAG-NAME>	Specifies the identifier for the LAG (link aggregation group).
<DIRECTION>	Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following:
`both`	Mirror both transmitted and received packets.
`rx`	Mirror only received packets.
`tx`	Mirror only transmitted packets.

Usage

There is a limit of source interfaces in each direction of a given mirror session:

Switch	Source interface limit per mirror session (4 possible sessions)
5420	64
6200	64

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.

When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface
  LAG-NAME      Enter a LAG name. For example, lag10
  PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2
switch(config-mirror-2)# source interface 1/1/3 tx
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-mirror-<SESSION-ID>`	Administrators or local user group members with execution rights for this command.

source vlan

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> {rx | tx | both}
```

Description

Mirroring with VLAN as a source is supported in the following traffic directions:

- **both** - traffic received and transmitted
- **rx** - only received traffic
- **tx** - only transmitted traffic

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs for a given mirror session. There is also a limit of 4096 source VLANs across all mirror sessions.

When changing a source VLAN in an enabled mirror session (i.e. adding, changing direction, or removing) mirrored packets being transmitted out of the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways.

- Reenter the **source vlan <VLAN-NUM> <direction>** command with the new preferred direction.
- Use the **no source vlan <VLAN-NUM> <direction>** form of the command with a direction (**rx** or **tx**) to selectively remove the specified direction.

Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Mirroring allows configuration of VLAN as a source. When VLAN source is configured in the **rx** direction, all packets are mirrored as they are received in the switch. When VLAN source is configured in **tx** direction, all packets are mirrored as they are transmitted out of the switch.

For packets bridged through the switch:

- If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in **rx** direction, packets are mirrored in the pre-routed form with the Destination MAC address as the switch address.
- If the mirror is configured in **tx** direction, packets are mirrored in post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in **both** direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in theingress and the egress packet processing pipeline. The following are the behavior for mirroring with VLAN as source:

- If the mirror is configured in the **rx** or **tx** direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the **both** direction, two copies of the packets are mirrored to the mirror destination.

The **no** form command will cease mirroring traffic from the specified source VLAN and remove the source from the mirror configuration.

Parameter	Description
`VLAN-NUM`	Selects the VLAN number.
`direction`	Specifies the direction of mirroring. **tx** (transmit), **rx** (receive), or **both**.

Examples

Creating a mirror session and adding a VLAN as a source of traffic in both directions on that port:

```
switch# configure terminal
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Creating a mirror session and adding two VLANs as sources of traffic:

directions:

```
switch# configure terminal
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Configuring the source in session 2 to receive by specifying the source interface configuration:

```
switch(config-mirror-2)# source vlan 10 rx
```

Removing the first source interface in session 2 entirely, and removing the transmit direction from the other so that mirroring only occurs in the receive direction:

```
switch(config-mirror-2)# source vlan 10 rx
switch(config-mirror-2)# source vlan 20 tx
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

Chapter 9

Monitoring a device using SNMP

Configuring SNMP: Refer to the *SNMP/MIB Guide* for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the *SNMP/MIB Guide* and specific information about the `show snmp trap` command to enable SNMP traps.

Chapter 10

Power-over-Ethernet

- The Power-over-Ethernet (PoE) subsystem manages power supplied to devices using standard Ethernet data cables. A Power Sourcing Equipment (PSE) supplies DC power as well as Ethernet connectivity to a Powered Device (PD) using a standard Ethernet cable. The maximum current depends on the PD Requested Class.
- A PoE subsystem contains two parts : a PSE and PD. A Power Sourcing Equipment (PSE) is a device that provides power through a standard Ethernet cable. A PoE capable switch functions as PSE. All Aruba PoE switches are considered as PSEs. A PD is a device powered by a PSE. Examples of PD are VoIP phones, Wireless APs, and IP cameras.
- When a PD or any network cable is connected to a PSE port, the PSE applies a detection voltage and measures the resistance value of the PD. If resistance is within IEEE 802.3 standard values (23 - 26k ohm), the connected device is treated as PD and classification begins. For legacy devices to be detected, you must enable prestandard detection on the switch.
- PDs are divided into different types and classes based on PD power requirements. The power supplied by the PSE is higher than the power PD draws to accommodate for the line losses that can result with the use of the standard maximum length cable(100m).
  - Type 1: PSE can supply maximum of 15.4W, and PD can draw a maximum of 13W.
  - Type 2: PSE can supply maximum of 30W, and PD can draw a maximum of 25.5W.
  - Type 3: PSE can supply maximum of 60W, and PD can draw a maximum of 51W.
  - Type 4: PSE can supply maximum of 90W, and PD can draw a maximum of 71W.
- Classes of PD:
  - Class 0: Type1 PD, it can draw a maximum of 13W.
  - Class 1: Type1 PD, it can draw a maximum of 3.84W.
  - Class 2: Type1 PD, it can draw a maximum of 6.49W.
  - Class 3: Type1 PD, it can draw a maximum of 13W.
  - Class 4: Type2 PD, it can draw a maximum of 25.5W.
  - Class 5: Type3 PD, it can draw a maximum of 40W.
  - Class 6: Type3 PD, it can draw a maximum of 51W.
  - Class 7: Type4 PD, it can draw a maximum of 62W.
  - Class 8: Type4 PD, it can draw a maximum of 71.3W.
- IEEE 802.3bt introduced 4-Pair PoE as a means of supplying higher power to PDs that need more than the current 25.5W supplied by IEEE 802.3at. To increase the available power without damaging the Ethernet cable, the standard introduced the ability to use all four pairs within the Ethernet cable instead of the two pairs used by previous standards (802.3at, 802.3af).
- Supported protocols:
  - Compatibility with IEEE 802.3af, 802.3at, 802.3bt and prestandard.
  - Long first class event supported on Type 3-4 PSE.
  - Support for Single Signature (SS) Type 0-6 and Dual Signature (DS) Type 0-4 PDs.
  - Multi-Event classification permits mutual ID of SS Class 0-6 and DS Class 0-4.

- Support LLDP Data Link Layer (DLL) Type 1-2 extension 12-octet TLV and Type 3-4 extension 29-octet TLV.
- Default PSE assigned class delivers the maximum PSE capable power at initial power up based on PD requested class.

- Always-on PoE is a feature that provides the ability for a switch to continue to provide power across user initiated reboots through software. Always-on PoE is enabled by default and no additional configuration is needed.

PDs only remain powered, no data transfer or PoE power negotiation can occur until the switch has completely booted up and in normal operation. PD faults occurring prior to full switch boot up will result in PoE power removal and restart the detection process only after switch returns to normal operation.

PoE commands

All PoE configuration commands except **quick-poe**, **threshold configuration** and **always-on poe configuration** are entered at the **config-if** context. The PoE threshold command is used at the system level whereas the **always-on poe** and **power-over-ethernet quick-poe** commands are set at the slot level. These commands can only be configured in the global configuration context.

lldp dot3 poe

```
lldp dot3 poe
no lldp dot3 poe
```

Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The **no** form of this command disables 802.3 TLV list in LLDP.

Examples

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

lldp med poe

```
lldp med poe [priority-override]
no lldp med poe [priority-override]
```

Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The **no** form of this command disables MED TLV list in LLDP.

Parameter	Description
`[priority-override]`	System defined name of the interface.

Examples

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp med poe
switch(config-if)# no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if)# lldp med poe priority-override
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

power-over-ethernet

```
power-over-ethernet
no power-over-ethernet
```

Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module.

The **no** form of this command disables per-interface power distribution.

Examples

Enabling per-interface power distribution:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# no power-over-ethernet
Interface PoE cannot be disabled when Quick PoE is enabled.
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

power-over-ethernet allocate-by

```
power-over-ethernet allocate-by {usage | class}
no power-over-ethernet allocate-by {usage | class}
```

Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The power allocation method can be changed on an interface through port-access (User roles or RADIUS). An allocation method when configured through port-access will replace the user configured method.

The **no** form of this command resets the action to default.

Parameter	Description
`usage`	Configures the usage-based allocation method.
`class`	Configures the class-based allocation method.

Usage

If you enable **pd-class-override** for an interface, the **allocate-by** configuration of that interface will be automatically changed to **class**. However, if you change the allocation method to **usage** when **pd-class-override** is still enabled, you will receive an error message stating that "The power allocation method cannot be changed when pd-class-override is enabled."

To remove **pd-class-override**, you can use the **no power-over-ethernet pd-class-override** command . It is important to note that **pd-class-override** requires the allocation method to be set to **class** and is enforced when configured through CLI. However, if you override the allocation method to **usage** via port-access, **pd-class-override** will not be in effect. Therefore, it is recommended that you do not override the allocation method to **usage** through port-access on interfaces configured with **pd-class-override**.

Examples

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

power-over-ethernet always-on

```
power-over-ethernet always-on <MODULE-ID>
no power-over-ethernet always-on <MODULE-ID>
```

Description

Always-on PoE is a feature that provides the ability to the switch to continue to provide power across a soft reboot. It is applicable only to the interfaces which were connected and delivering before the soft reboot. Also, power will not be delivered if power to the switch is interrupted. This command enables or disables the always-on PoE feature at the switch or the slot level. By default, always-on PoE is enabled at the switch or the slot level.

The **no** form of this command disables power distribution on soft reboot.

Parameter	Description
<MODULE-ID>	Module number to apply always-on PoE configuration.

Examples

Enabling per-interface power distribution:

```
switch(config)# power-over-ethernet always-on 1/1
```

Disabling per-interface power distribution:

```
switch(config)# no power-over-ethernet always-on 1/1
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

power-over-ethernet assigned-class

```
power-over-ethernet assigned-class {3 | 4 | 6}
no power-over-ethernet assigned-class
```

Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the sybsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The **no** form of this command resets the action to default.

Examples

Setting PoE assigned class:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# interface 1/1/1
switch(config)# power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

power-over-ethernet pre-std-detect

```
power-over-ethernet pre-std-detect
no power-over-ethernet pre-std-detect
```

Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. Aruba switches are backward compatible and will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The **no** form of this command resets the action to default.

Examples

Enabling pre-standard device detection:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling pre-standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet priority

```
power-over-ethernet priority {critical | high | low}
no power-over-ethernet priority {critical | high | low}
```

Description

Sets PoE priority for an interface Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

power-over-ethernet quick-poe

```
power-over-ethernet quick-poe <MODULE-ID>
no power-over-ethernet quick-poe <MODULE-ID
```

Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The **no** form of this command disables quick PoE.

Parameter	Description
<MODULE-ID>	Specifies module number for quick PoE configuration .

Examples

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/2
```

```
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
```

```
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

power-over-ethernet threshold

```
power-over-ethernet threshold <PERCENTAGE>
no power-over-ethernet threshold <PERCENTAGE>
```

Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The **no** form of this command resets the action to default.

Parameter	Description
<PERCENTAGE>	Excess power consumption trap threshold. Range 1-99.

Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution rights for this command.

power-over-ethernet trap

```
power-over-ethernet trap
no power-over-ethernet trap
```

Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if)# no power-over-ethernet trap
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution rights for this command.

show lldp local

`show lldp local-device [<INTERFACE-ID>]`

Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: **member/slot/port**

Examples

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
===============

Port-ID          : 1/1/10
Port-Desc        : "1/1/10"
Port VLAN ID     : 0

PoE Plus Information

PoE Device Type    : Type 2 PSE
Power Source       : Primary
Power Priority     : low
PSE Allocated Power: 25.0 W
PD Requested Power : 25.0 W
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor

`show lldp neighbor [<INTERFACE-ID>]`

Description

Displays detailed information about a particular neighbor connected to a particular interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: **member/slot/port**

Examples

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10

Port                             : 1/1/10
Neighbor Entries                 : 1
Neighbor Entries Deleted         : 0
Neighbor Entries Dropped         : 0
Neighbor Entries Aged-Out        : 0
Neighbor Chassis-Name            : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description     : ArubaOS (MODEL: 325), Version HPE_ANW IAP
Neighbor Chassis-ID              : 84:d4:7e:ce:5d:68
Neighbor Management-Address      : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled     :
Neighbor Port-ID                 : 84:d4:7e:ce:5d:68
Neighbor Port-Desc               : eth0
TTL                              : 120
Neighbor Port VLAN ID            :
Neighbor PoEplus information     : DOT3
Neighbor Device Type             : TYPE2 PD
Neighbor Power Priority          : Unkown
Neighbor Power Source            : Primary
Neighbor Power Requested         : 25.0 W
Neighbor Power Allocated         : 0.0 W
Neighbor Power Supported         : No
Neighbor Power Enabled           : No
Neighbor Power Class             : 5
Neighbor Power Paircontrol       : No
Neighbor Power Pairs             : SIGNAL
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show power-over-ethernet

```
show power-over-ethernet [member <MEMBER-ID>] [brief]
```

Description

Displays the status information of the full system.

Parameter	Description
<MEMBER-ID>	Displays the detailed status of given member.
<IFNAME>	Display the detailed status of given port.
brief	Display the brief status of all ports or the given port.

Examples

Showing sample output for show power-over-ethernet on standalone box with VSF capabiity:

```
switch# show power-over-ethernet

System Power Status for member 1

  Configured  Power Status     : No redundancy
  Operational Power Status     : No redundancy
  Total Available Power        : 740 W
  Total Failover Pwr Avl       :   0 W
  Total Redundancy Power       :   0 W
  Total Power Drawn            :   0 W +/- 6W
  Total Power Reserved         :   0 W
  Total Remaining Power        : 740 W
  Trap Threshold               : 80 %
  Trap Enabled                 : Yes
  Always-on PoE Enabled        : 1/1
  Quick PoE Enabled            : None


Internal Power
        Total Power
  PS     (Watts)        Status
  ----- -------------  ---------------------
  1     0              Absent
  2     740            Ok

System Power Status for member 2

  Configured  Power Status     : No redundancy
  Operational Power Status     : No redundancy
  Total Available Power        : 600 W
  Total Failover Pwr Avl       :   0 W
  Total Redundancy Power       :   0 W
  Total Power Drawn            :   0 W +/- 6W
  Total Power Reserved         :   0 W
  Total Remaining Power        : 600 W
  Trap Threshold               : 80 %
  Trap Enabled                 : Yes
  Always-on PoE Enabled        : None
  Quick PoE Enabled            : None


Internal Power
        Total Power
  PS     (Watts)        Status
  ----- -------------  ---------------------
```

```
  1     0               Absent
  2     600             Ok
```

Showing sample output for power-over-ethernet member:

```
switch# show power-over-ethernet member 1

System Power Status for member 1

  Configured  Power Status     : No redundancy
  Operational Power Status     : No redundancy
  Total Available Power        : 740 W
  Total Failover Pwr Avl       :   0 W
  Total Redundancy Power       :   0 W
  Total Power Drawn            :   0 W +/- 6W
  Total Power Reserved         :   0 W
  Total Remaining Power        : 740 W
  Trap Threshold               : 80 %
  Trap Enabled                 : No
  Always-on PoE Enabled        : 1/1
  Quick PoE Enabled            : 1/1

Internal Power
        Total Power
  PS    (Watts)         Status
  ----- -------------  ---------------------
  1     0               Absent
  2     740             Ok
```

Showing sample output for power-over-ethernet brief in a VSF stack:

```
switch# show power-over-ethernet brief

Status and Configuration Information for PoE

  Member 1 Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
    Quick PoE Enabled: None

PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port     PD     Cls Type
Port     En  Priority Detect  Act   Rsrvd   Draw   Status       Sign
-------  --- ------   ------- ----- ------  ------ ---------    -----  --- ----
1/1/1    Yes Low      Off     Class  0.0 W   0.0 W Denied       None   4   2
1/1/2    Yes Critical Off     Usage  1.6 W   1.5 W Delivering*  Single 0   1
1/1/3    Yes High     Off     Class 54.0 W  25.5 W Delivering*^ Dual   1/3 3
1/1/4    No  Low      On      Usage  0.0 W   0.0 W Disabled     None   N/A N/A

  Member 2 Power Status
    Available: 600 W  Reserved: 0.00 W  Remaining: 600 W
    Always-on PoE Enabled: None
    Quick PoE Enabled: None

PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port     PD     Cls Type
Port     En  Priority Detect  Act   Rsrvd   Draw   Status       Sign
-------  --- ------   ------- ----- ------  ------ ---------    -----  --- ----
2/1/1    Yes Low      Off     Class  0.0 W   0.0 W Searching    None   N/A  N/A
2/1/2    Yes Critical Off     Usage  0.0 W   0.0 W Searching    None   N/A  N/A
2/1/3    Yes High     Off     Class  0.0 W   0.0 W Searching    None   N/A  N/A
```

```
2/1/4    No  Low      On      Usage  0.0 W   0.0 W Disabled     None   N/A  N/A

*This port may go down in the event of a PSU failure.
^This port is power demoted due to user config or power availabilty.
```

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief

Status and Configuration Information for port 1/1/1

  Member 1Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port     PD     Cls Type
Port     En  Priority Detect  Act   Rsrvd   Draw   Status       Sign
-------  --- ------   ------- ----- ------  ------ ---------    -----  --- ----
1/1/1    Yes Low      Off     Class  0.0 W   0.0 W Denied       None   4   2
```

Showing sample output for power-over-ethernet brief for interface range:

For 6300 Switch series:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief

Status and Configuration Information for port 1/1/1-1/1/2

  Member 1Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
    Quick PoE Enabled: None
PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port     PD     Cls Type
Port     En  Priority Detect  Act   Rsrvd   Draw   Status       Sign
-------  --- ------   ------- ----- ------  ------ ---------    -----  --- ----
1/1/1    Yes Low      Off     Class  0.0 W   0.0 W Denied       None   4   2
1/1/2    Yes Critical Off     Usage  1.6 W   1.5 W Delivering*  Single 0   1
```

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief

Module 1/3 is not physically present.
```

Showing sample output for power-over-ethernet brief for a missing member:

```
switch# show power-over-ethernet member 3 brief

Member 3 is not physically present.
```

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1

Interface 2/1/1 is not present.
```

For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Chapter 11
Aruba AirWave

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the *AOS-CX Release Notes*.

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
```

```
snmp-server vrf mgmt
!
...
```

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface.
	Firmware versions
	Displays neighbor devices connected to AOS-CX switches
	Device topology
Configuration management	Partial configuration
Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the switch by entering the `snmp-server vrf` command.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

switch(config)# snmpv3 context Admin
```

   For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

   SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as `mgmt`. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as `default` or `mgmt`.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

AirWave commands

logging

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME>} [ {udp [<PORT-NUM>] }|{tcp
[<PORT-NUM>} | {tls [<PORT-NUM> [auth-mode {certificate|subject-name}] [legacy-tls-
renegotiation]}] [severity <LEVEL>] [vrf <VRF-NAME>] [include-auditable-events]
[filter <FILTER-NAME>] [ rate-limit-burst <BURST> [rate-limit-interval <INTERVAL>] ]

no logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>
```

Description

Enables syslog forwarding to a remote syslog server.

The `no` form of this command disables syslog forwarding to a remote syslog server.

Parameter	Description
{*<IPV4-ADDR>* | *<IPV6-ADDR>* | *<HOSTNAME>*}	Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.
[udp [*<PORT-NUM>*] | tcp [*<PORT-NUM>*]]	Specifies the UDP port or TCP port of the remote syslog server to receive the forwarded syslog messages.
udp [*<PORT-NUM>*]	Range: 1 to 65535. Default: 514
tcp [*<PORT-NUM>*]	Range: 1 to 65535. Default: 1470
tls [*<PORT-NUM>*]	Range: 1 to 65535. Default: 6514
include-auditable-events	Specifies that auditable messages are also logged to the remote syslog server.
severity *<LEVEL>*	Specifies the severity of the syslog messages: ■ `alert`: Forwards syslog messages with the severity of `alert (6)` and `emergency (7)`. ■ `crit`: Forwards syslog messages with the severity of `critical (5)` and above. ■ `debug`: Forwards syslog messages with the severity of `debug (0)` and above. ■ `emerg`: Forwards syslog messages with the severity of `emergency (7)` only. ■ `err`: Forwards syslog messages with the severity of `err (4)` and above ■ `info`: Forwards syslog messages with the severity of `info (1)` and above. Default. ■ `notice`: Forwards syslog messages with the severity of `notice (2)` and above. ■ `warning`: Forwards syslog messages with the severity of `warning (3)` and above.
auth-mode	Specifies the TLS authentication mode used to validate the certificate. ■ certificate: Validates the peer using trust anchor certificate based authentication. Default. ■ subject-name: Validates the peer using trust anchor certificates as well as subject-name based authentication.
legacy-tls-renegotiation	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.
vrf *<VRF-NAME>*	Specifies the VRF used to connect to the syslog server. Optional. Default: `default`

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of `err (4)` and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF `lab_vrf`:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject name
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

snmp-server community

```
snmp-server community <STRING>
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

Parameter	Description
<STRING>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

snmp-server host

```
snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The `no` form of this command removes the specified trap/inform receiver.

Configuring `snmpv3 informs` is not supported.

Parameter	Description
`<IPv4-ADDR>`	Specifies the IP address of a trap receiver in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. You can remove leading zeros. For example, the address `192.169.005.100` becomes `192.168.5.100`.
`trap version <VERSION>`	Specifies the trap notification type for SNMPv1 or v2c. Available options are: `v1` or `v2c`.
`inform version v2c`	Specifies the inform notification type for SNMPv2c.
`trap version v3`	Specifies the trap notification type for SNMPv3.

Parameter	Description
`user <NAME>`	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.
`community <STRING>`	Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: `public`.
`<UDP-PORT>`	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.
`vrf <VRF-NAME>`	Specifies the name of the VRF on which to send the notifications.

Examples

```
switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port
2000
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

snmp-server vrf

```
snmp-server vrf <VRF-NAME>
no snmp-server vrf <VRF-NAME>
```

Description

Configures the VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF.

The `no` form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<VRF-NAME>	Specifies the VRF on which the SNMP agent listens for incoming requests. The SNMP agent can listen on either the `mgmt` or `default` VRF. If configured for both, the SNMP agent listens on `default`, which has a higher priority.

Example

```
switch(config)# snmp-server vrf default
```

```
switch(config)# no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

snmpv3 context

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The `no` form of this command removes the specified SNMP context.

Parameter	Description
<NAME>	Specifies the name of the context. Range: 1 to 32 printable ASCII

Parameter	Description
	characters, excluding space and question mark (?).
vrf *<VRF-NAME>*	Specifies the VRF associated with the context. Default: default.
community *<STRING>*	Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user

```
snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass {plaintext | ciphertext}
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass {plaintext | ciphertext} <PRIV-PWORD>] ]
no snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass <PRIV-PWORD>] ]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context.

The `no` form of this command removes the specified SNMPv3 user.

Parameter	Description
<NAME>	Specifies the SNMPv3 username. Range 1 - 32 printable ASCII characters, excluding space and question mark.
auth *<AUTH-PROTOCOL>*	Specifies the authentication protocol used to validate user logins. Available options are: `md5` or `sha`.
auth-pass {plaintext | ciphertext} *<AUTH-PWORD>*	Specifies the SNMPv3 user password. Range for `plaintext` is 8 - 32 printable ASCII characters, excluding space and question

Parameter	Description
	mark. Range for `ciphertext` is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.
`priv <PRIV-PROTOCOL>`	Specifies the SNMPv3 security protocol (encryption method). Available options are: `aes` or `des`.
`priv-pass {plaintext | ciphertext} <PRIV-PWORD>`	Specifies the SNMPv3 user privacy passphrase. Range for `plaintext` is 8 - 32 printable ASCII characters, excluding space and question mark. Range for `ciphertext` is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

Examples

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Removing an SNMPv3 user named `Admin`:

```
switch(config)# no snmpv3 user Admin
```

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Copying an SNMP user from switch 1 to switch 2.

On switch 1, configure a user called **Admin**, then issue the `show running-config` command to display switch configuration settings. The snmpv3 user command uses the `ciphertext` option to protect the users's passwords.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des priv-pass plaintext myprivpass
switch1(config)# exit
switch1# show running-config
Current configuration:
```

```
!
!Version AOS-CX TL.10.00.0003-8017-gdeb0606~dirty
!
!
!
snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
ssh server vrf mgmt
!
!
!
!
interface mgmt
    no shutdown
    ip dhcp
vlan 1
```

On switch 2, execute the snmpv3 user command that was displayed by `show running-config` on switch 1. This creates the user on switch 2 with the same configuration settings.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass
ciphertextAQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=priv
des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

Chapter 12
Support and Other Resources

Accessing Aruba Support

Aruba Support Services	https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html
AOS-CX Switch Software Documentation Portal	https://arubanetworking.hpe.com/techdocs/AOS-CX/help_portal/Content/home.htm
Aruba Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-650-750-0350 (Backup—Toll Number)
International telephone	https://www.hpe.com/psnow/doc/a50011948enw

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

Aruba Hardware Documentation and Translations Portal	https://arubanetworking.hpe.com/techdocs/hardware/DocumentationPortal/Content/home.htmm
Aruba software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End-of-Life information	https://networkingsupport.hpe.com/end-of-life

Accessing Updates

You can access updates from the Aruba Support Portal at https://networkingsupport.hpe.com.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe./notifications/subscriptions (requires an active Aruba Support Portal account to manage subscriptions). Security notices are viewable without an Aruba Support Portal account.

Warranty Information

To view warranty information for your product, go to https://www.arubanetworks.com/support-services/product-warranties/.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see https://www.arubanetworks.com/company/about-us/environmental-citizenship/.

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